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RESEARCH ARTICLE

Vehicle Routing and Crew Assignment Problem With the Consideration of Time-Dependent Travel and Service Time

ANTONIA ILABACA[1], GERMÁN PAREDES-BELMAR[1], AND BRODERICK CRAWFORD[2]
[1]School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Brasil 2241, Valparaíso 2362807, Chile
[2]School of Informatics Engineering, Pontificia Universidad Católica de Valparaíso, Valparaíso 2362807, Chile

Corresponding author: Germán Paredes-Belmar (german.paredes@pucv.cl)

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ABSTRACT In this paper, we present the Time-Dependent Vehicle Routing Problem with Fleet and Crew Assignment (TD-VRPFCA), which aims to design efficient delivery routes in urban environments considering Time-Dependent Travel Times (TD-TT) for the road network and Time-Dependent Service Times (TD-ST) for in-building deliveries.The problem also involves assigning heterogeneous crews, composed of drivers, delivery personnel, and multifunctional delivery personnel, who can drive and deliver. To solve the TD-VRPFCA, we propose a Mixed-Integer Linear Programming (MILP) formulation for small-size instances and Black Widow Optimization (BWO) metaheuristic for medium and large-scale cases. The search intensification procedures. Computational experiments demonstrate that for small instances, the BWO obtains the same solutions as the MILP model with a fraction of the computational time, while for large instances it outperforms Genetic Algorithm and Tabu Search approaches according to Friedman and Nemenyi statistical tests, providing better fitness values and lower CPU times. Finally, a real-world case study based on an urban delivery company in Santiago, Chile, confirms the applicability of the proposed approach for realistic routing and crew planning under variable traffic conditions. The results show that the TD-VRPFCA provides an effective framework for supporting last-mile logistics decisions with complex workforce configurations.

INDEX TERMS City logistics, crew assignment, time-dependency, vehicle routing, black widow optimization algorithm.

I. INTRODUCTION

The Vehicle Routing Problem (VRP) is a combinatorial optimization problem that aims to find optimal routes for a fleet of vehicles to serve a set of customers with known demands [1]. The VRP has been extended in various ways to account for realistic constraints in logistics operations. In particular, the increasing complexity of last-mile logistics in urban areas has motivated the incorporation of time windows, stochastic travel times, time-dependency, dynamic customer requests, and multi-level service operations [1], [2].

These extensions are especially pertinent in the context of Urban Logistics, which introduces a set of distinctive operational challenges. Urban environments are characterized by traffic congestion, limited parking infrastructure, high customer density, and in-building customers, which are factors that significantly impact the efficiency of delivery operations. Moreover, urban deliveries span both Business-to-Customer (B2C) and Business-to-Business (B2B) contexts, each with unique logistical requirements. B2C deliveries typically involve small orders and frequent trips, often with a high number of delivery points per route, whereas B2B deliveries tend to consist of fewer stops with larger shipment volumes, frequently destined

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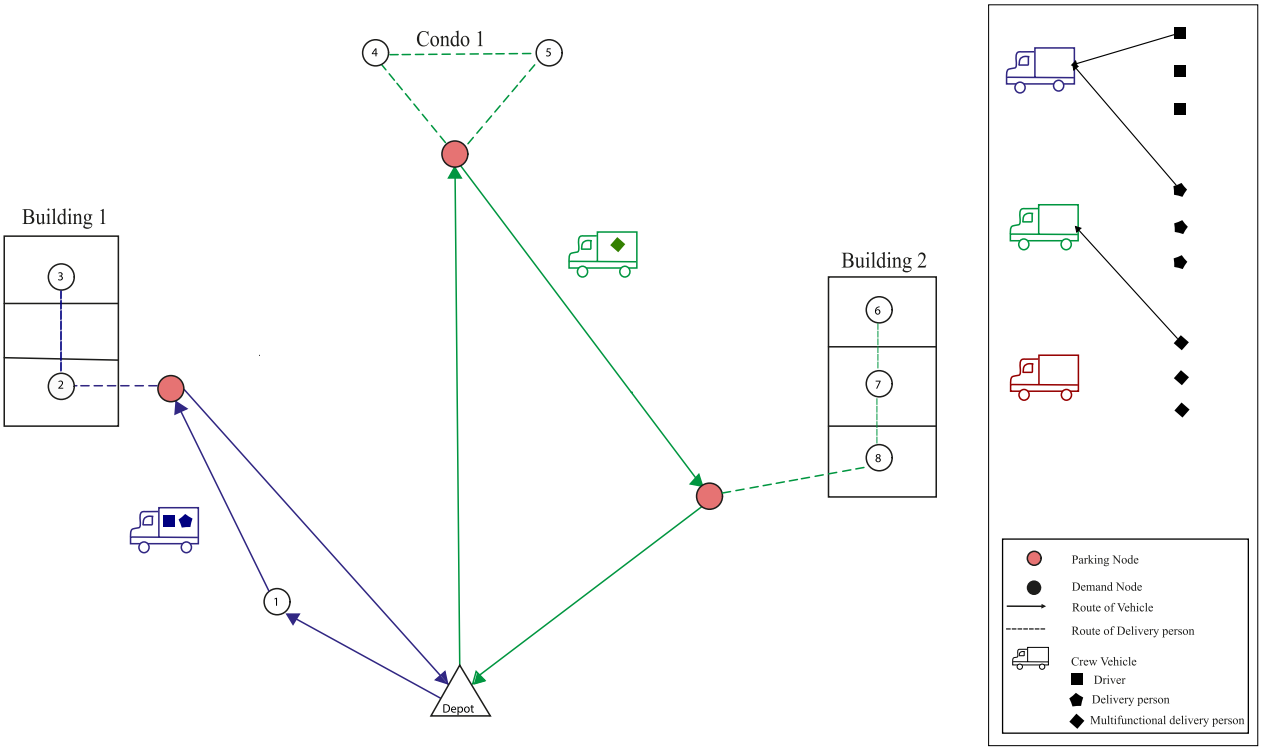


FIGURE 1. Example VRPFCA.

for commercial buildings or warehouses [3]. In both cases, achieving operational efficiency depends not only on optimal routing but also on effective crew coordination and the ability to address location-specific constraints [4], [5].

In response to these challenges, this paper introduces, models and solves the Time-Dependent Vehicle Routing Problem with Fleet and Crew Assignment (TD-VRPFCA). This problem integrates several critical and interdependent dimensions relevant to B2B and B2C delivery systems in urban environments. First, it considers time-dependent travel times (TD-TT), which allow the model to capture the variability of street-level traffic conditions during different periods of the day. Second, it incorporates time-dependent service times (TD-ST), which represent the time variability within buildings, such as elevator congestion or shared infrastructure, that affects building deliveries. Third, it introduces a flexible crew assignment component with heterogeneous worker roles, including drivers, delivery personnel, and multifunctional delivery personnel (MFDPs) who are capable of both driving and delivering. The combination of these three aspects contributes to a novel and realistic solution procedure that reflects the operational decisions and constraints facing logistics providers.

The TD-VRPFCA scheme is shown in Figure 1, where the central depot (triangle) is the vehicle point of origin and return (represented in blue and green icons), and the customers or demand nodes are represented with circles. At the bottom of the figure, three vehicles, three drivers, three delivery personnel, and three multifunctional delivery personnel are available. In this case, one driver and one delivery person are assigned to the blue vehicle, while one multifunctional delivery person is assigned to the green vehicle. Figure 1 also indicates that the first vehicle route, marked by blue arrows, handles deliveries to customers or demand nodes 1, 2, and 3 with a crew of one driver and one delivery person. Customer 1 can be directly accessed by the vehicle, while customers 2 and 3 are within Building 1. As a result, the vehicle parks outside Building 1, and one DP completes the in-building deliveries at customers' doors. In contrast, the second vehicle (green arrows) delivers products to customers 4, 5, 6, 7, and 8 with a crew of just one MFDP. Customers 4 and 5 are located in areas inaccessible to the vehicle or where easy delivery is possible on foot, such as in a condo, for example, Condo 1. The vehicle parks and deliveries are made on foot by the MFDP. Meanwhile, customers 6, 7, and 8 are located within building 2, and the MFDP meets the in-building deliveries.

The objective of this research is to minimize the total distribution time by jointly optimizing vehicle routes, delivery schedules, and crew assignments under realistic urban characteristics. To achieve this, we propose:

1) A Mixed-Integer Linear Programming (MILP) formulation for solving small instances of the TD-VRPFCA optimally;

2) a Black Widow Optimization (BWO) metaheuristic adapted to efficiently solve large-scale, real-world instances; and
3) an extensive set of computational experiments, including benchmark tests and a case study based on a Chilean urban delivery company, to validate the proposed methodology.

Although each of these modeling components, time-dependent travel times and service times, and flexible crew composition, have been considered separately in previous works [6], [7], [8], this is, to our knowledge, the first study to integrate them into a unified routing and crew assignment framework. When treated separately, these factors often result in underestimated travel and service durations, as well as suboptimal crew allocation. By jointly modeling their interactions, our integrated framework overcomes these limitations and enhances both accuracy and efficiency.

This paper is organized as follows. Section II presents a literature review with related problems to the TD-VRPFCA. Section III details the solution methodology, describing the mathematical model and the BWO metaheuristic. Section IV summarizes the computational experiences and discusses the results. Finally, Section V presents the conclusions derived from the study.

II. LITERATURE REVIEW

In this section, we detail the related approaches to our problem, focusing on three areas: (i) routing problems in urban logistics with time dependency and crew assignment, (ii) solution methods for complex VRP variants, and (iii) models involving in-building delivery.

A. VEHICLE ROUTING IN URBAN AREAS

The Vehicle Routing Problem (VRP) is a well-known combinatorial optimization problem with a long history in logistics and operations research [1]. It seeks to determine the most efficient set of routes for a fleet of vehicles to serve a set of customers, thereby optimizing an objective function (e.g., cost, time, or number of vehicles). Over time, several extensions of the VRP have been proposed to capture operational realities such as vehicle capacities, time windows, split deliveries, multiple depots, and heterogeneous fleets [9]. In urban settings, additional constraints are often necessary due to traffic congestion, limited parking, and densely populated customer locations.

A significant stream of research has focused on time-dependent travel times, resulting in the Time-Dependent Vehicle Routing Problem (TDVRP). In this variant, travel time between locations depends on the time of departure, capturing real-world traffic patterns. Seminal works such as those by Ichoua et al. in [10] and Ehmke in [11] introduced and analyzed formulations and algorithms that handle dynamic travel times and street congestion in urban environments. More recently, hybrid time-dependent models have been proposed for both planning and real-time routing [12]. Additionally, Xiao and Konak in [13] addressed the heterogeneous green vehicle routing and scheduling problem under time-varying traffic congestion. Meanwhile, Blauth et al. in [14] provided a comprehensive overview of vehicle routing with time-dependent travel times, encompassing both theory and practice, as well as benchmarks. These works have significantly advanced our understanding of time-dependent routing. However, they generally assume static service processes and omit the effect of workforce composition on route duration.

Large-scale urban operations exemplify how demand forecasting and operational coordination are utilized to manage highly time-dependent transportation systems. For instance, the *Delivering London 2012* special issue reports on transport demand forecasting and operations delivery for the Olympic Games [15], as well as programme assurance and legacy aspects [16], [17]. These practical studies underline the operational importance of accurate time-dependent estimates and coordinated resource deployment, complementing the modeling focus of the present TD-VRPFCA.

Another key research direction involves incorporating workforce assignment and labor roles into the VRP. Early studies considered fixed crews per vehicle, but do not explore the interaction between crew composition and service efficiency. The Vehicle Routing Problem with Multiple Deliverymen (VRPMD) is one such extension, where the number of delivery workers affects the duration of the service [7]. Munari and Morabito in [18] extended this by modeling team-based deliveries, showing how crew configuration impacts route duration and cost. More flexible models enable the dynamic assignment of workers to vehicles and consider heterogeneous roles, such as drivers and delivery personnel with distinct capabilities [19]. While these approaches incorporate human resources into routing, they typically assume static traffic conditions and do not account for time-dependent travel.

In healthcare logistics, Zhang et al. in [20] proposed a model that integrates vehicle routing with manpower planning for non-emergency ambulance transfer services, explicitly incorporating staff availability and patient service time constraints. Similarly, Kovacs et al. in [21] addressed the service technician routing and scheduling problem, emphasizing synchronization between vehicles and personnel, a crucial aspect in services involving skilled staff such as nurses. In the airline catering industry, Ho and Leung in [22] tackled manpower scheduling problems using metaheuristics, highlighting the importance of efficient crew assignment in time-sensitive delivery operations. These examples underline the importance of considering human resources in routing problems. However, they are sector-specific and do not generalize well to dense urban last-mile delivery.

In the context of dense cities, some approaches model hybrid delivery systems such as the Park-and-Loop Routing Problem (PLRP), where drivers park at central locations and delivery workers complete local deliveries on foot [23],

[24]. Similarly, two-echelon and pedestrian routing models have been studied in urban logistics [25]. However, these works typically assume a fixed separation between driver and delivery roles and often overlook the time dependency in travel or service durations.

A particularly relevant yet underexplored feature is the inclusion of multi-functional delivery personnel (MFDPs), who are logistics workers responsible for last-mile deliveries. This flexibility is increasingly adopted in practice, particularly in companies that utilize agile delivery schemes and the gig economy labor model. Despite its practical relevance, few models account for both crew heterogeneity and operational flexibility, particularly in environments where access conditions to buildings affect service time and feasibility.

In the same urban context, in-building deliveries introduce operational challenges that differ from street-level routing, including vertical trips, elevator congestion, restricted access, and limited internal navigation infrastructure [8]. Gökalp and A. Uğur in [26] optimized delivery routes for autonomous or human agents in multi-floor buildings using metaheuristics, while Kim et al. in [27] studied delivery flows in office buildings considering entry, delivery, and exit stages. Chen et al. in [28] incorporated vertical travel time into a 3D VRP framework, optimizing last-mile delivery routes to customers located inside buildings. Although these contributions offer insight into building-aware logistics, most models focus on localized internal routing and are not integrated into city-scale frameworks. Furthermore, they rarely consider crew assignment or heterogeneous worker roles, and none explicitly account for the combined effects of traffic congestion, service times, and in-building dynamics. Therefore, the literature on building delivery provides useful building blocks, but leaves open the challenge of integration with urban vehicle routing.

B. SOLUTION METHODS FOR INTEGRATED ROUTING AND CREW PROBLEMS

Solving VRP extensions that incorporate time dependency, service heterogeneity, and crew assignment poses significant computational challenges. Exact methods, particularly Mixed-Integer Linear Programming (MILP) models, have been widely used to solve small- and medium-sized instances of VRP variants [6], [29]. These formulations are capable of expressing complex constraints and allow optimality guarantees, but are generally limited by scalability issues.

To address large-scale problems, the literature offers a wide variety of metaheuristics and hybrid approaches. Genetic Algorithms (GA), Iterated Local Search (ILS), Tabu Search, and Large Neighborhood Search (LNS) are among the most prominent, due to their flexibility in dealing with realistic operational constraints [37], [38], [40]. Some of these approaches have been successfully adapted to problems involving time-dependent travel, dynamic demand, or multi-echelon structures [39], [42].

Recently, Black Widow Optimization (BWO) has emerged as a promising population-based metaheuristic inspired by the natural selection behavior of black widow spiders [43]. It has been applied to problems such as vehicle routing with profits [44] and nurse scheduling [45], showing competitive results. However, no previous study has applied BWO to a problem with the combined complexity of time-dependent routing, building-aware service times, and flexible crew assignment.

In this paper, we present a MILP formulation for small-scale instances of the TD-VRPFCA and develop a BWO-based metaheuristic tailored to the problem's structure, including encoding schemes for route and crew assignment, a feasibility repair mechanism, and local search intensification. The TD-VRPFCA integrates city-scale routing with realistic service times for deliveries to customers located both at street level and inside buildings. It incorporates time-dependent service times for in-building deliveries, capturing elevator congestion, vertical delays, and the influence of delivery crew configurations, building on the empirical functions reported in Paredes-Belmar et al. in [8]. By incorporating these variable service times, the TD-VRPFCA accounts for essential aspects of in-building logistics while maintaining a computationally tractable model suitable for city-level optimization.

Table 1 summarizes the most relevant related articles, indicating whether they address street-level routing (VR), crew assignment (CA), multi-functional delivery personnel, time-dependent travel times (TD-TT), time-dependent service times (TD-ST), and the solution method used. As shown, most works concentrate on one or two dimensions, but none jointly integrate all five.

In summary, the main contribution of this study is to extend the state of the art in the VRP literature by jointly modeling and solving three critical dimensions that have only been addressed separately in prior research: (i) time-dependent travel times, capturing traffic variability in urban networks; (ii) time-dependent service times, explicitly accounting for in-building complexities such as elevator congestion and vertical movements; and (iii) flexible crew assignment, incorporating heterogeneous and multifunctional delivery personnel who can both drive and deliver. This integration enables a more realistic representation of urban delivery systems, where traffic congestion and the configuration of delivery personnel and vehicle fleet interact in complex ways.

III. METHODOLOGY

We present two approaches to solving the TD-VRPFCA problem. The first considers a MILP formulation to address all requirements of the problem. This formulation is planned to solve small-size instances. The second is a BWO metaheuristic for solving instances of real-world size. Note that the TD-VRPFCA problem is NP-hard, as it is a generalization of the classic TDVRP, which is also NP-hard. For this reason, approximate methods such as heuristics and metaheuristics are required. The objective of using the BWO metaheuristic

TABLE 1. Summary of related literature on vehicle routing, crew assignment, and in-building delivery problems.

Reference	VR	CA	MFDP	TD-TT	TD-ST	Solution Method
Vehicle Routing with Crew Assignment						
[7]	✓	✓				Branch & Cut, ACO, Tabu Search
[30]	✓	✓				GRASP, Local Search
[31]	✓	✓				ILS, LNS
[19]	✓	✓				Branch-and-cut Algorithm
[18]	✓	✓				Branch-and-price-and-cut
[32]	✓	✓				Multi-Objective EA
[25]	✓	✓				Branch-and-Cut Algorithm
[23]	✓	✓				TSP heuristics
[24]	✓	✓				SLNS
Manpower Routing and Crew Scheduling Problems						
[20]	✓	✓				MILP, Heuristics
[21]	✓	✓				Adaptive Methods
[22]		✓				Metaheuristics
Time-Dependent Travel Time (TD-TT) VRP						
[33]	✓			✓		MILP Model
[6]	✓			✓		MILP Model
[34]	✓			✓		Simulated Annealing
[35]	✓			✓		LANTIME Heuristic
[11]	✓			✓		TDNNDP, TDIHDP Heuristics
[4]	✓			✓		TDIHDP, LANTIME Heuristics
[36]	✓			✓		VNS
[37]	✓			✓		Markov Decision Model, ADP
[38]	✓			✓		MILP, GRASP, ILS
[39]	✓			✓		MILP, Evolutionary Algorithm
[40]	✓			✓		MILP, ILS
[29]	✓			✓	✓	MILP, NSGA II
[41]	✓			✓	✓	MILP, Robust model, exact method
Current Work	✓	✓	✓	✓	✓	MILP, Black Widow Optimization

is to reduce CPU time compared to traditional models in real-world instances.

A. THE MILP MODEL

The TD-VRPFCA considers a complete directed graph $G = (N, A)$ where N is a set of nodes and $A$is the set of arcs. Let $I \subset N$ denote the set of vehicle stops (e.g., buildings or street-level addresses). The depot is represented by a starting node o and a dummy return node s, both corresponding to the same physical location. A fleet of K vehicles is available, each operated with one of the D possible crew combinations composed of workers of types C

The planning horizon is divided into a set L of discrete time intervals [10], with street-level vehicle speed between nodes $(i, j) \in N$ during interval $l \in L$ given by v_{ij}^{l}.

Service time t_{id}^{l} depends on the delivery type at node i. For building deliveries, the service time is determined by the crew configuration $d \in D$, the number of customers or floor servers, and the building's traffic intensity during time intervals $l \in L$. For street-level customers, service time depends only on the assigned crew combination. A detailed list of sets, parameters, and decision variables is provided in Table 2.

The model is as follows:

$$\min \; t_{\text{MAX}} \tag{1}$$

$$\text{subject to} \sum_{i \in N \setminus \{o\}} \sum_{l \in L} x_{oi}^{kl} \leq 1 \quad \forall k \in K \tag{2}$$

$$\sum_{i \in N \setminus \{s\}} \sum_{l \in L} x_{is}^{kl} \leq 1 \quad \forall k \in K \tag{3}$$

$$\sum_{k \in K} z_i^k = 1 \quad \forall i \in I \tag{4}$$

$$\sum_{i \in N \setminus \{j,s\}} \sum_{k \in K} \sum_{l \in L} x_{ij}^{kl} = 1 \quad \forall j \in N \setminus \{o\} \tag{5}$$

$$\sum_{i \in N \setminus \{j,o\}} \sum_{k \in K} \sum_{l \in L} x_{ji}^{kl} = 1 \quad \forall j \in N \setminus \{s\} \tag{6}$$

$$\sum_{i \in N \setminus \{j,s\}} \sum_{l \in L} x_{ij}^{kl} = \sum_{i \in N \setminus \{j,o\}} \sum_{l \in L} x_{ji}^{kl}$$
$$\forall k \in K, j \in N \setminus \{o, s\} \tag{7}$$

$$\sum_{d \in D} y_d^k \leq 1 \quad \forall k \in K \tag{8}$$

$$\sum_{d \in D} y_d^k \geq z_i^k \quad \forall k \in K, i \in N \tag{9}$$

$$\sum_{j \in N \setminus \{i,o\}} \sum_{l \in L} x_{ij}^{kl} \leq M_1 z_i^k \quad \forall k \in K, i \in N \tag{10}$$

$$w_i^k \leq M_2 z_i^k \quad \forall k \in K, i \in N \tag{11}$$

$$\sum_{i \in N} q_i z_i^k \leq \sum_{d \in D} Q^k y_d^k \quad \forall k \in K \tag{12}$$

$$\sum_{d \in D} \sum_{k \in K} \omega_d^c y_d^k \leq V^c \quad \forall c \in C \tag{13}$$

$$\sum_{d \in D} \sum_{c \in C} \omega_d^c y_d^k \leq W^k \quad \forall k \in K \tag{14}$$

TABLE 2. Sets, parameters, and decision variables for the TDVRPFCA MILP model.

Notation	Description
I	set of vehicle stops (customers/buildings)
o, s	depot nodes
N	set of nodes: $N = I \cup \{o, s\}$
A	set of arcs: $A = \{(i,j) : i,j \in N, i \neq j\}$
K	set of vehicles
D	set of crew combination
L	set of time intervals
C	set of worker types
S_{ij}^{l}	speed for each time interval $l \in L$ between each pair of nodes $i \in N, j \in N \setminus \{i\}$
δ_{ij}	distance between each pair of nodes: $i \in N, j \in N \setminus \{i\}$
T_{id}^{l}	service time for each customer $i \in I$ served with crew combination $d \in D$ in the time interval $l \in L$
ω_{d}^{c}	number of workers type $c \in C$ in each crew combination $d \in D$
W^{k}	maximum crew number for each vehicle $k \in K$
V^{c}	maximum number of workers $c \in C$ in a crew
q_i	demand for each node $i \in N : q_o = q_s = 0$
Q^{k}	capacity for each vehicle $k \in K$
Ll^{l}	lower limit time interval $l \in L$
Ul^{l}	upper limit time interval $l \in L$
M	Big-M. $M_1 = \|N\|$ and $M_2 = \max_{l \in L}\{Ul^{l}\}$
x_{ij}^{kl}	binary decision variable. 1, if the vehicle $k \in K$ travel from node $i \in N$ to node $j \in N \setminus \{i\}$ in the time interval $l \in L$; 0, otherwise.
y_{d}^{k}	binary decision variable. 1, if the vehicle $k \in K$ use the crew combination $d \in D$. 0, otherwise.
z_{i}^{k}	binary decision variable. 1, if the vehicle $k \in K$ visits the node $i \in N$; 0, otherwise.
w_{i}^{k}	continuous decision variable. Arrival time to node $i \in N$ with the vehicle $k \in K$
t_{MAX}	continuous decision variable. Maximal time to arrival at depot

$$w_j^k \geq w_i^k + \left(\frac{\delta_{ij}}{S_{ij}^l}\right) x_{ij}^{kl} + T_{id}^l y_d^k - M_2(1 - x_{ij}^{kl})$$
$$\forall k \in K, i \in N \setminus \{s\}, j \in N \setminus \{i, o\}, d \in D, l \in L \quad (15)$$

$$w_i^k \leq w_j^k + \left(\frac{\delta_{ij}}{S_{ij}^l}\right) x_{ij}^{kl} + T_{id}^l y_d^k - M_2(x_{ij}^{kl} - 1)$$
$$\forall k \in K, i \in N \setminus \{s\}, j \in N \setminus \{i, o\}, d \in D, l \in L \quad (16)$$

$$w_o^k = Ll^l x_{oj}^{kl} \quad \forall k \in K, j \in N \setminus \{o\}, l \in L \quad (17)$$

$$Ll^l x_{ij}^{kl} \leq w_i^k \quad \forall k \in K, i \in N, j \in N \setminus \{i\}, l \in L \quad (18)$$

$$Ul^l \geq w_i^k - M_2(1 - x_{ij}^{kl}) \quad \forall k \in K, i \in N, j \in N \setminus \{i\}, l \in L, M_2 \gg 0 \quad (19)$$

$$T_{\mathrm{MAX}} \geq w_i^k - w_o^k \quad \forall k \in K, i \in N \setminus \{o\} \quad (20)$$

$$x_{ij}^{kl}, y_d^k, z_i^k \in [0, 1] \quad \forall k \in K, i \in N, j \in N \setminus \{i\}, d \in D, l \in L \quad (21)$$

$$t_{\mathrm{MAX}}, w_i^k \geq 0 \forall k \in K, i \in N \quad (22)$$

The objective function (1) minimizes the maximum time to arrival at the depot for all vehicles and crews t_{MAX}, which accounts for both travel and service time. It corresponds to minimizing the makespan, a classical objective in scheduling and routing literature. The makespan has been widely used in job shop scheduling, particularly when synchronization of resources or shift planning is critical [1], [46]. The relevance of makespan-based objectives has been documented in routing contexts where vehicle return time and shift coverage are key performance indicators [9], [47]. Therefore, minimizing t_{MAX} provides an optimization objective aligned with operational goals in real-world urban distribution systems.

Constraints (2)-(7) ensure a connected route sequence for each vehicle from and to the depot. Constraints (2) and (3) specify that each vehicle performs at most one route sequence, and it starts and returns to the same depot. Constraints (4)-(6) guarantee that each node is visited by a single vehicle within a time interval. Finally, constraints (7) establish flow balance for each node and vehicle.

Constraints (8) represent that each vehicle has exactly one crew combination. Constraints (9) ensure that if vehicle k uses a crew combination d, this vehicle must visit a customer i. Constraints (10) ensure that if the vehicle sequence k visits a customer i before a node j, the vehicle must deliver products at node i. Constraints (11) relate arrival time and visit variables w and z, where if the vehicle k arrives at node i at w time, then this node is marked as visited.

Constraints (12)-(14) state maximum vehicle and crew capacities. Constraints (12) indicate that each vehicle has a limited capacity Q^k. Constraints (13) indicate that each crew can have at most V^c workers of type c. Constraints (14) assure that each vehicle can transport a limited number of workers.

Constraints (15)-(17) count the arrival time of each vehicle at each visited node. These equations ensure that the arrival time takes into account the previous arrival time, the travel time to the current node, and the service time at the previous node. These constraints also prevent sub-tours in the routes. Constraints (18) and (19) update the time intervals used based on their lower limits. Constraints (20) count the t_{MAX} value, which is related to the objective function.

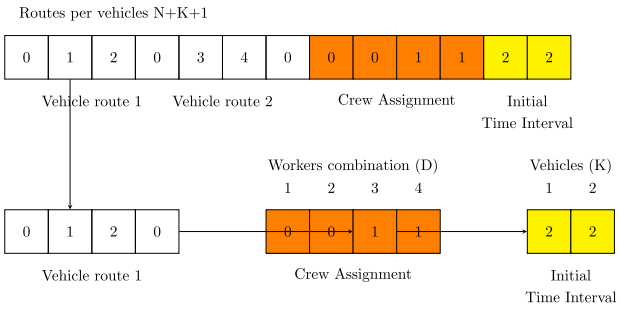


FIGURE 2. Solution representation.

Finally, constraints (21)-(22) define the domain of decision variables.

B. NOVEL BLACK WIDOW OPTIMIZATION FOR TIME-DEPENDENT VEHICLE ROUTING PROBLEM WITH FLEET AND CREW ASSIGNMENT

The Black Widow Optimizer (BWO) algorithm is a metaheuristic introduced by Hayyolalam and Kazem in [43] is a population-based metaheuristic that simulates evolutionary mechanisms to explore and exploit the solution space. Its search process is organized into three phases: (1) procreation, (2) cannibalism, and (3) mutation, which collectively balance diversification and intensification.

The BWO has demonstrated competitive performance in combinatorial and continuous optimization problems, including scheduling, feature selection, cloud computing, Internet of Things, cybersecurity, route design, and networking [42], [44], [45], [48], [49], [50], [51], [52]. To our knowledge, this is the first work to use the BWO for a routing and crew assignment problem.

The selection of BWO is motivated by two factors. First, unlike more established heuristics (e.g., GA or Tabu Search), BWO naturally incorporates a strong diversification mechanism (cannibalism and mutation) that reduces the risk of premature convergence. Second, it has been successfully adapted to structurally similar problems to the TD-VRPFCA, demonstrating robustness in terms of combinatorial complexity.

The original BWO is based on the mating behavior of black widow spiders and is used for population-based optimization. It starts with an initial population of spiders, each representing a feasible routing and crew assignment plan. The algorithm has three evolutionary phases: procreation, cannibalism, and mutation, all of which mimic the behavior of these spiders. Thus, the BWO is designed, adapted, and implemented to solve the VRPFCA. This algorithm incorporates a constructive heuristic for calculating fitness, which is equivalent to the objective function for the MILP model.

1) SOLUTION REPRESENTATION AND POPULATION INITIALIZATION

To solve the Time-Dependent Vehicle Routing Problem with Crew Assignment (TD-VRPFCA), each individual in the population, referred to as a ''widows'', represents a complete routing and crew assignment plan. The main decision components include the sequence of customer visits for each vehicle, the crew assigned to each vehicle, and the time interval when each route starts.

The solution is encoded as a concatenated of three parts (Figure 2). The first segment represents the customer sequence for each vehicle, with zeros indicating returns to the depot. This part has a length of $N + K + 1$, ensuring that each customer is visited once and vehicle capacity constraints are satisfied.

The second segment, shown in orange in Figure 2, represents the crew assignment. Each position corresponds to a vehicle $k \in K$ and indicates which delivery crew $d \in D$ is assigned to it. Crew groups are ordered according to their capacity, ensuring that the assignment respects heterogeneity and vehicle crew compatibility.

The third segment (in yellow) defines the initial time interval of each vehicle, with values ranging from 1 to L. This enables the algorithm to capture variations in travel and service times throughout the planning horizon. For instance, in Figure 2 vehicle 1 begins its route in the time interval 2.

From this representation, the decision variable of the MILP model, x_{ij}^{kl}, y_d^k, and z_i^k can be directly inferred. For example, $x_{01}^{12} = 1$, $y_3^1 = 1$, and $z_1^1 = 1$ in the illustrated case. These values are subsequently used to compute other dependent variables such as the starting service time w_i^k and the total time t_{MAX}, using the fitness function.

Feasibility is checked against the problem constraints. When a violation occurs, such as exceeding vehicle capacity or an unassigned customer, a repair mechanism reassigns customers or adjusts crew composition to restore feasibility while maintaining route integrity.

The fitness function evaluates the total time of each solution. This evaluation uses a constructive heuristic specifically developed for the TD-VRPFCA, which reproduces the MILP objective structure. The heuristic computes the time-dependent travel and service durations for each vehicle and crew pair based on δ_{ij}, v_{ij}^l, and t_{id}^l. The procedure is detailed in Algorithm 1. Notably, the crew assignment variable y_d^k is extracted from the encoded solution. This hybridization between the BWO and the constructive heuristic constitutes one of the methodological contributions of this work.

After the population is initialized and each individual's fitness is evaluated, the algorithm proceeds to the procreation phase.

2) PROCREATION PHASE

The procreation phase is responsible for generating new candidate solutions by recombining information from pairs of parent individuals. In the proposed adaptation of BWO, pairs of parents are randomly selected from the population according to a predefined reproduction rate (np). Each individual encodes the sequence of customer crew assignment

Algorithm 1 Fitness Calculation

Require: Solution, δ_{ij}, v^l_{ij}, t^l_{id}, N, K, D, Ll^l, Ul^l, y^k_d
Ensure: Fitness value
Initialize: $k \leftarrow 1; l \leftarrow$ solution$[|N|+k+|D|+1]; w \leftarrow \{\}$; $T_{\text{MAX}} \leftarrow 0$
for it $\in \{0, 1, \ldots, |N| + |K| - 1\}$ **do**
Select i, j from solution
if $i = o$ and $j \neq o$ **then**
$w^k_o \leftarrow Ll^l$; $w^k_j \leftarrow w^k_o + \delta_{ij}/v^l_{ij}$
Update time intervals if needed
else if $i \neq o$ and $j \neq o$ **then**
$w^k_j \leftarrow w^k_i + \sum_{d \in D}(t^l_{id} \cdot y^k_d) + \delta_{ij}/v^l_{ij}$
Update time intervals if needed
else if $i \neq o$ and $j = o$ **then**
$w^k_s \leftarrow w^k_i + \sum_{d \in D}(t^l_{id} \cdot y^k_d) + \delta_{is}/v^l_{is}$
$k \leftarrow k + 1; l \leftarrow$ solution$[|N| + k + |D| + 1]$
end if
end for
$T_{\text{MAX}} \leftarrow \max\{w^k_s \mid k \in K\}$
return T_{MAX}

and starting time intervals, as described in the previous subsection.

To produce a new generation of children, two crossover points are randomly chosen within the routing segment of the solution string. The customer sequences between these points are exchanged between parents to form two new children. After the crossover, a route correction procedure ensures that all customers are visited exactly once, eliminating any duplicates or omissions that might occur during the exchange.

The crew assignment and initial time interval segments are also exchanged between the parents to preserve consistency between the vehicles and their corresponding crews. As a result, two offspring are generated from each parent pair. Both children share the same routing structure, but differ in the assigned crew inherited from their respective parents. This mechanism maintains route integrity while diversifying the workforce configuration. That procedure is illustrated in Figure 3.

3) CANNIBALISM PHASE

At this stage, there are three types of cannibalism. The first is sexual cannibalism, sibling cannibalism, and parental cannibalism. In the original BWO, these mechanisms represent successive selection processes in which weaker individuals are eliminated, allowing only the strongest solutions to survive. The strength of each individual is determined by its fitness value, which in this problem corresponds to the total time.

In the proposed adaptation, sexual cannibalism refers to the selection between parent pairs, where the individual with lower fitness (i.e., higher total time) is removed from the population. Sibling cannibalism corresponds to the comparison among offspring generated in the procreation phase, retaining only those with better performance. Finally, parental cannibalism compares the best offspring with their parent, keeping the stronger solution to maintain population quality.

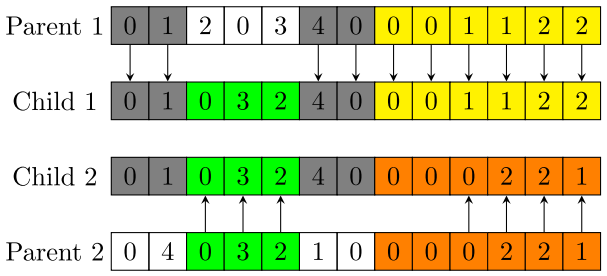


FIGURE 3. Procreation representation.

This hierarchical elimination process follows the same logic as the standard BWO but is evaluated based on the structure of the TD-VRPFCA. Consequently, the population gradually evolves toward feasible and efficient routing and crew configurations, ensuring that only the most promising solutions survive to the next generation.

4) MUTATION PHASE

The mutation stage is a crucial step in improving the population and increasing its diversity. Several individuals are selected for mutation in this stage, and two elements within their arrays are randomly changed. The new array's fitness is then calculated to determine whether it should be included in the population.

In the novel BWO, the mutation process is based on a genetic algorithm, where solution elements are randomly selected and exchanged. The time is then calculated to determine whether this change leads to an improvement. The new array is kept in the population if the improvement is maintained. However, if the improvement is not sustained, it is discarded.

It is essential to note that this mutation process is not performed in every iteration. Instead, it is applied with a given probability (e.g., 20%) to reduce computational workload and avoid excessive disruption of high-quality solutions while still promoting population diversity.

These stages are summarized in the Algorithm 2.

IV. COMPUTATIONAL EXPERIENCE

This section aims to evaluate the performance of the proposed solution approach through two main components: (i) synthetic instances to assess algorithmic efficiency and (ii) a real-case scenario to analyze its practical applicability. Section IV-A presents the synthetic benchmark design, Section IV-B reports algorithm performance results, and Section IV-C discusses the application of the model to a real last-mile delivery case. All TD-VRPFCA instances were executed on a computer with AMD Ryzen(7) 5700G, Radeon Graphics, a clock speed of 3.8 GHz, and 32 GB of RAM. The

Algorithm 2 Black Widow Optimization (BWO) for TD-VRPFCA

Require: n, np, cr, nm, $NPOP$, $I_{\max}$
Ensure: Near-optimal solution for TD-VRPFCA
Initialize population pop of $NPOP$ black widows
for $I = 1$ to $I_{\max}$ **do**
Sort pop by fitness (ascending)
Select top np solutions into pop_1
Procreation and Cannibalism
for $i = 1$ to np **do**
Randomly select two parents from pop_1
Generate offspring from procreation
Eliminate weak parent
With probability cr, eliminate some offspring (incl. parent if needed)
Save survivors into pop_2
end for
Mutation
for $i = 1$ to nm **do**
Select and mutate a widow from pop_1
Save new solution into pop_3
end for
Update Population
$pop \leftarrow pop_2 \cup pop_3$
Retain best solution found
end for
return Best solution from pop

MILP model was coded in Python 3.12 and solved using the Gurobi 10.0.2 commercial solver. The BWO metaheuristic was also programmed in Python 3.12.

The computational experiments employed both synthetic and real-world datasets. The real dataset corresponds to an urban delivery study in Santiago, Chile, provided by a last-mile logistics company. The dataset includes customer locations, vehicle capacity, and time-dependent travel and service time. Data preprocessing ensured consistency and completeness, missing travel times were interpolated from adjacent intervals, and travel speeds were discretized into predefined time-dependent intervals ($l \in L$). This structure allows the algorithm to capture realistic traffic dynamics and crew composition.

The complete set of benchmark instances, real case data, and source code are publicly available to ensure reproducibility of the results in the GitHub repository: https://github.com/antoilabaca/TD-VRP-FCA

A. SMALL INSTANCE

Since this TD-VRPFCA is a novel problem, the literature does not present suitable test instances to solve it. Thus, test cases were designed to compare and assess the performance of the MILP and BWO models. Specifically, in the small instance, the objective is to validate BWO with the MILP model and to compare the efficiency of both in terms of computational time.

A small instance was created using the information presented in [53]. The study focused on analyzing the parcel market by tracking all the movements of six delivery personnel in London, UK. The paper details travel times, the buildings and floors visited, service times, and the demand for each building. It also provides information on vehicle capacities and the number of personnel involved in deliveries.

Variations in travel and service times were randomized over 10 hours. Each vehicle was operated by a crew consisting of one driver and one delivery person. On average, the total delivery time was 5 hours and 10 minutes, with 1 hour and 33 minutes spent driving and the remaining time spent parked (including loading/unloading operations). Additionally, a delivery person spent approximately 3 hours and 36 minutes delivering packages to various buildings.

The evaluation of this instance has two main objectives: (i) to assess the performance of the Novel BWO in comparison with the optimal results obtained using the MILP, and (ii) to evaluate how crew assignment and time dependencies affect the planning of routes and travel times, using a real-world point of comparison.

Figure 4 presents the original delivery plan for 15 customers located on different floors of ten buildings (nodes) during a workday from 9:30 am to 2:00 pm. In particular, Figure 4a), based on [53], was built by tracking a vehicle and a delivery person walking with an altimeter. Figure 4b) presents a detailed sequence of vehicles for the ten nodes.

TABLE 3. Description of feasible crew combinations.

Crew (D)	Dr	DP	$MFDP$
1	1	1	0
2	1	2	0
3	1	3	0
4	1	4	0
5	0	0	1
6	0	1	1
7	0	2	1
8	0	3	1

The small instance introduces time dependency, as deliveries are performed in dense urban areas. Ten time intervals of 60 minutes are considered, with a speed varying between 5 and 20 km/h, depending on the time of day. The instance includes eight possible crew combinations, with a limit of 5 workers per vehicle, a capacity of 600 packages, and a fleet of two vehicles described in Table 3, where I indicates the number of buildings/stops, the number of packages per building/stop is randomly drawn from a uniform distribution in the range of 6 to 60, Dr indicates the available number of drivers, DP indicates the available number of delivery persons and $MFDP$ sets the available number of multi-functional persons. The instance was executed in the Gurobi solver with a time limit of one hour. In contrast, we run the BWO metaheuristic with a stopping condition of 10 iterations. Table 4 displays the results we obtained for two to ten nodes, where CPU(s) is the total CPU time, in seconds, for each run, TT (h) indicates the travel time, in hours, required for

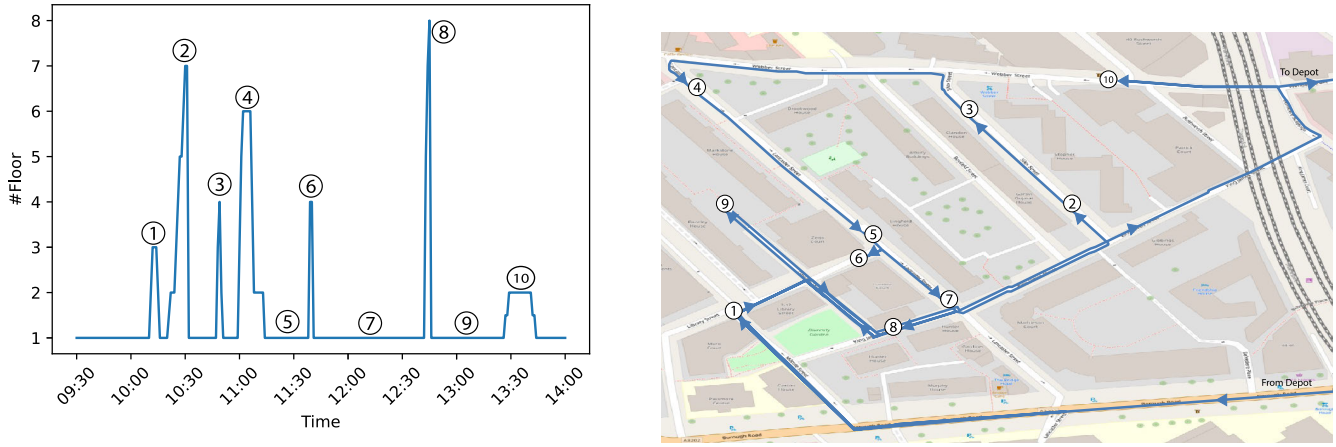


FIGURE 4. Route of vehicle in a workday (a) Customers and floors visited by a delivery person (b) Vehicle sequence for visiting ten buildings in south bank, london, UK.

all deliveries, and GAP indicates the Integrality GAP of the MILP model.

In this scenario, the BWO metaheuristic provides the same objective value as the MILP model in the first seven instances. However, in the eight-node instance, the MILP model did not find the optimal solution within the time limit. Furthermore, the GAP increased exponentially, from 0.3% to 83.74%. However, the BWO provided feasible solutions in less than 2 seconds in all cases, using an initial population of 50 spiders and 10 iterations.

Regarding the results of crew composition, the two vehicles were used consistently, which explains the time difference between the Allen et al. [53] case and the others. To obtain a fair comparison and similar result to [53], a test instance was solved with one vehicle and only one multi-functional person. The results are displayed in Figure 5, and it is evident that the times are similar, with 3 hours and 53 minutes.

Finally, Figure 6 presents a convergence graph for a small instance involving ten buildings and one truck. We can observe that BWO converges quickly within the first ten iterations, highlighting the efficiency of this meta-heuristic in this particular case. In this instance, we can conclude that the BWO metaheuristic is crucial for operational decisions, as the MILP model fails to find high-quality feasible solutions with more than ten nodes. In contrast, the novel BWO consistently produces equal or better feasible solutions in a short time. Furthermore, the use of different crew combinations leads to a reduction in delivery time, underscoring the importance of planning both the crew and routes concurrently.

These results highlight the computational advantages of the BWO metaheuristic compared to the MILP model and demonstrate its scalability. As the number of nodes increases, the MILP model consistently reaches its time limit without guaranteeing optimality, resulting in a gap of up to 83.7%. In contrast, BWO maintains low CPU times while producing high-quality solutions, showcasing its potential for large-scale, real-world urban delivery problems.

Moreover, the flexibility in crew composition is a valuable aspect of the modeling process. The ability to assign different types of personnel (#Dr, #DP, #MFDP) allows for more efficient route planning. This is particularly evident in cases where the MILP and BWO solutions have the same objective value but differ in crew configurations. This suggests that operational efficiency can be enhanced not only through routing decisions but also through crew design.

B. LARGE INSTANCES

The construction of large instances is based on the instances of Balseiro et al. in [54], which provides information for 100 nodes, including their demand and available vehicles. This information is derived from Solomon in [55] and a time-dependent matrix for these nodes based on Ichoua et al. in [10]. Thus, 24 instances are solved using the same worker data as the smaller instances (Table 3), applying the BWO metaheuristic. The objective of these test instances is to observe the performance of Novel BWO on larger instances, comparing it with the Genetic Algorithm (GA) and Tabu Search metaheuristic (TS). Consequently, BWO is based on the well-known Genetic Algorithm (GA), as noted in [43], and since Tabu Search is not a population-based metaheuristic, we allow its comparison with other types of metaheuristics. Tabu Search has been particularly recognized for its efficiency in Vehicle Routing Problems [56], and for many years it has remained one of the most widely used and effective metaheuristics for this class of problems [57], [58]

The problem considers symmetric networks of 100 nodes, as shown in Figure 7, where the demand between 10 and 50 units and the capacity of vehicles vary. There are 25 vehicles available, each with a capacity between 200 to 1,000 units. The instances cover three time intervals-morning peak hours, the middle of the day (lunchtime), and evening peak hours- along with a symmetric category matrix. Each

TABLE 4. Small instance results.

I	MILP			BWO		Crew Composition		
	CPU (s)	TT (h)	GAP	CPU (s)	TT (h)	#D	#DP	#MFDP
2	<1.00	1.16	0.0%	<1.00	1.16	1	1	1
3	<1.00	1.46	0.0%	<1.00	1.46	1	1	1
4	<1.00	1.47	0.0%	<1.00	1.47	0	2	2
5	3.33	1.77	0.0%	1.01	1.77	1	3	1
6	8.65	1.77	0.0%	1.15	1.77	2	2	0
7	179.92	2.06	0.0%	1.30	2.06	2	4	0
8	3,600	2.07	0.3%	1.42	2.07	0	3	2
9	3,600.	2.37	74.2%	1.49	2.37	0	4	2
10	3,600	2.40	83.7%	1.55	2.38	1	2	1

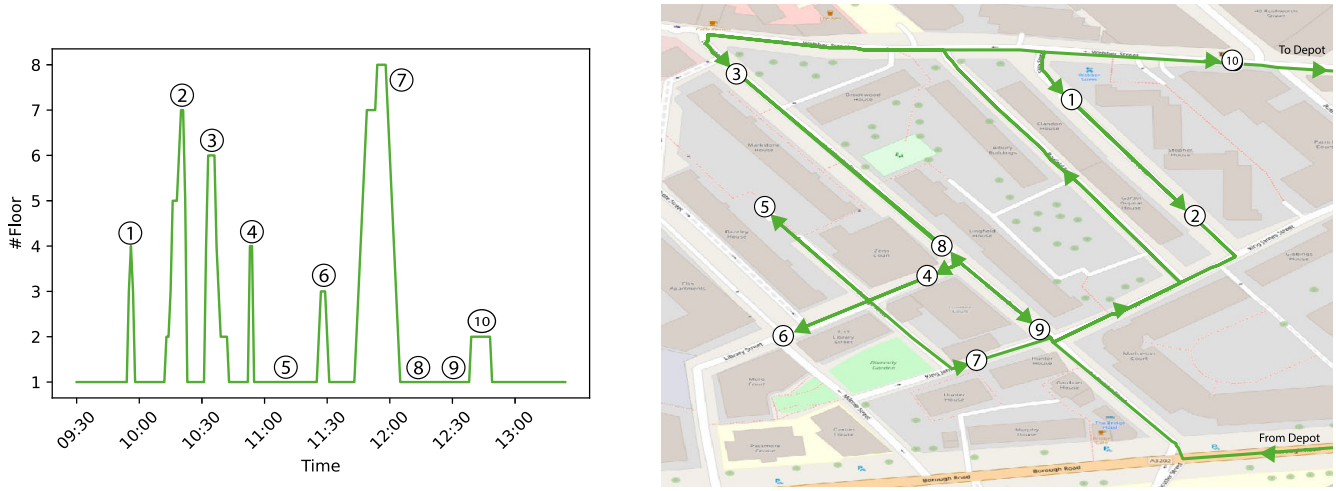


FIGURE 5. Route of the vehicle in a workday with BWO (a) Floors visited (b) Vehicle sequence for the ten-building instance.

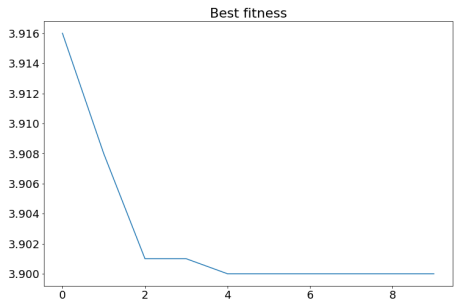


FIGURE 6. Convergence graph.

category is associated with a distinct step speed function; Category 1 has the slowest speed, while Category 3 has the fastest speed between nodes.

We assume that higher traffic conditions on the streets lead to increased congestion at the nodes (e.g., greater worker exit flow), resulting in longer service times. The time-dependent service time was not originally included in the test instances; however, it has been estimated based on time intervals, the category of speed function, and crew composition, utilizing the methodology described in [8].

The instances are categorized into clustered (RC101, RC201) and non-clustered or randomized (R101, R201). In this work, they are named according to the name_tdmatrix_category. For example, RC201_1_1 refers to instance RC201 with time-dependency Matrix 1 and Category 1.

Since this is a larger instance, it is necessary to tune the parameters of both metaheuristics. The parameters used in the process were carefully analyzed. As previously mentioned, the GA and the original BWO share a similar composition; the main difference lies in the cannibalism stage of the BWO, which is claimed to improve the performance of the GA [43]. To verify this, the same solution representation, constructive heuristic, and parameters presented previously for both are used. Parameter selection was performed using the BWO and applied to both algorithms to ensure consistent conditions and facilitate comparison, thereby justifying the use of the BWO. The TS parameters setting is different because it is a solution-based metaheuristic and has other performance and parameters with respect to BWO and GA this algorithm changes their efficiency with respect to the maximal iteration of the algorithm.

The parameters involved in our study are as follows: *np* (rates of procreation), *cr* (rates of survival against cannibalism, specific to the BWO algorithm), *nm* (rate of mutation), *NPOP* (population size), and I_{max} (maximum number of iterations). For larger instances, *NPOP* and I_{max} have a significant impact on the quality of the solution. This means that increasing these two parameters generally leads to

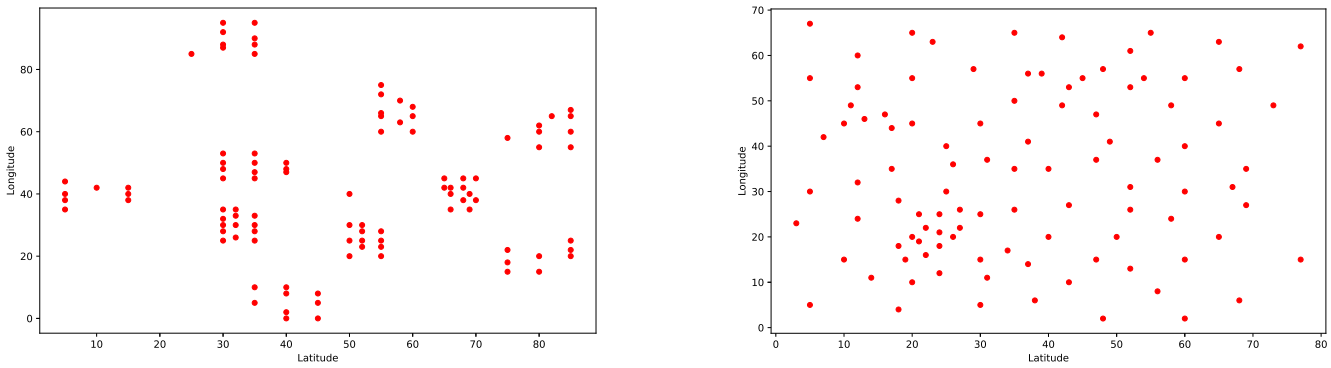


FIGURE 7. Nodes large instance (a) Clustered instances (b) No-clustered instances.

better solutions each time the algorithm is executed. However, larger values for these parameters also increase computation time, substantially raising CPU usage. Therefore, careful consideration is required when tuning these parameters.

To optimize the performance of the BWO algorithm, we conducted experiments with various parameter combinations and utilized the Friedman non-parametric test to statistically compare the improvements in solutions across different configurations. For this analysis, we executed the BWO algorithm 20 times for 4 of 24 instances (RC101_1_1, RC201_1_1, R101_1_1, R201_1_1) using the following parameter values: *np* at 40% and 60%, *cr* at 50% and 100%, *nm* at 5% and 10%, *NPOP* at 20, 50, and 100 individuals, and I_{max} at 50 and 100 iterations. For TS has only one critical parameter and is I_{max} at 100, 500, and 1000 iterations. These parameter choices are based on literature concerning similar cases.

After conducting the Friedman test, the best parameter combination identified was a 60% rate of procreation, a 10% rate of mutation, a 50% rate of survival against cannibalism, a population size of 20 individuals, and a maximum of 100 iterations. The BWO and GA used the same parameters, and TS used 1000 iterations in I_{max}. A provides a summary of the results of the analyses for BWO and TS.

Notice that the MILP model can provide feasible solutions for 100-node instances within one hour of computation, but with an average integrality GAP of around 100%. Thus, we do not report those MILP results. However, metaheuristic algorithms yield solutions in less than an hour.

In our comparison between BWO, GA, and TS, we used the Friedman non-parametric test. This test is commonly applied when comparing more than two algorithms, especially when dealing with repeated measurements on the same instances, and it does not assume normality in the data. We executed each instance 20 times to obtain representative samples and compared the main results in Table 5. We conducted the Friedman test on a data matrix, where each row represents a different instance and each column corresponds to a different algorithm (BWO, GA, and TS). For our analysis, we calculated the average for each instance.

Initially, we conducted a test to evaluate the fitness value of each algorithm. The Friedman statistic is 15.75, with a *p*-value of 0.0004, indicating a statistically significant difference between the algorithms in terms of their fitness values. Subsequently, we applied a post-hoc Nemenyi analysis to elucidate the differences among the algorithms further, and the results are presented in the Table 6

From the Table 5, we observe that the BWO metaheuristic demonstrates the best performance in terms of fitness value. The post-hoc analysis indicates that this difference is significant when compared to TS; however, it is not significant in relation to GA. This is understandable, as the primary objective of BWO is to achieve a good solution in less CPU time compared to GA.

Next, we conducted the Friedman test concerning CPU time. In this case, the Friedman statistic is 42.46, and the p-value is 0.00001, indicating a significant difference between the algorithms in terms of CPU time. The results of the post-hoc Nemenyi analysis for CPU time are presented in Table 7.

The post-hoc analysis of CPU time reveals a significant difference among the three algorithms. Notably, the difference between TS and GA is more pronounced, as illustrated in Table 5. Additionally, the difference in CPU time between BWO and GA is also significant, indicating that while BWO achieves similar results in fitness value compared to GA, it does so in less time.

We present a diagram illustrating the efficiency (Average rank) of BWO in comparison to a GA and TS, demonstrating that BWO has a good performance when compared to traditional algorithms like GA or TS, as shown in Figure 8.

Finally, Figure 9 displays the results for the RC101_2_2 instance with 100 nodes at each iteration to show the speed at which it converges to the solution. It is evident that the best result is achieved around the 60th iteration in this case.

These findings highlight that BWO not only improves GA and TS in terms of computational efficiency but also provides a reliable and scalable alternative for solving large instances, even under time-dependent travel conditions and diverse crew configurations. Therefore, BWO emerges as a promising tool

TABLE 5. Comparison of BWO, GA, and TS performance on benchmark instances.

Instance	BWO				GA				TS			
	Fitness	SD	CPU (s)	SD	Fitness	SD	CPU (s)	SD	Fitness	SD	CPU (s)	SD
RC101_1_1	371.6	59.8	22.9	3.4	510.3	27.8	34.6	2.7	590.3	351.37	23.0	1.8
RC101_2_1	599.1	56.0	23.1	4.0	588.4	7.5	37.2	3.6	716.9	351.07	20.7	0.4
RC101_3_1	834.3	73.6	21.3	3.2	871.5	34.8	36.0	2.3	933.4	232.91	21.2	1.1
RC101_1_2	385.4	92.9	23.2	3.7	377.1	37.7	35.7	2.1	676.3	349.71	21.0	1.5
RC101_2_2	606.0	59.9	22.5	3.7	593.3	22.5	36.8	3.4	898.2	210.59	21.1	0.4
RC101_3_2	864.5	72.2	21.1	2.6	888.3	27.8	35.4	2.4	815.9	216.78	21.0	0.4
RC201_1_1	403.1	50.6	24.8	3.6	429.3	43.9	34.3	3.0	576.8	334.85	20.5	0.5
RC201_2_1	564.7	92.5	22.2	2.1	567.6	78.2	35.8	2.9	807.8	194.86	20.8	0.7
RC201_3_1	837.8	75.2	23.3	3.4	914.9	37.0	32.5	1.2	817.3	118.36	21.1	0.3
RC201_1_2	428.8	62.8	22.7	2.1	451.2	75.1	34.4	2.0	869.7	224.04	21.2	0.3
RC201_2_2	590.5	70.8	23.8	3.8	586.6	32.8	34.9	2.1	769.6	181.00	21.1	1.0
RC201_3_2	848.3	98.9	23.1	3.3	938.4	98.6	32.8	3.2	889.5	139.81	21.0	0.3
R101_1_1	531.0	11.1	25.5	2.5	536.4	11.6	39.3	2.6	598.1	329.65	25.5	6.0
R101_2_1	699.7	26.9	27.9	4.2	706.8	27.2	42.5	7.3	791.1	359.48	24.3	5.2
R101_3_1	1015.3	26.1	26.4	3.6	1013.0	23.1	37.6	1.3	758.4	202.87	28.2	8.4
R101_1_2	535.7	25.7	28.0	4.2	541.8	14.6	38.0	3.1	594.4	413.30	28.1	5.2
R101_2_2	696.9	23.7	26.3	3.6	697.4	23.6	36.9	3.0	829.4	250.69	25.3	5.4
R101_3_2	1033.5	16.4	29.1	4.1	1033.2	9.8	37.5	2.6	910.2	229.79	26.4	4.4
R201_1_1	261.3	26.2	22.8	3.3	308.5	25.9	34.6	2.5	843.4	370.89	20.8	0.7
R201_2_1	403.9	58.6	23.9	3.6	372.3	38.9	32.4	2.4	736.2	299.50	20.9	1.2
R201_3_1	738.3	51.1	23.0	3.3	770.2	37.1	31.4	2.2	863.5	205.21	20.4	0.2
R201_1_2	245.3	40.5	25.1	3.9	269.6	26.1	32.0	2.7	718.6	264.52	20.6	0.6
R201_2_2	398.7	54.6	22.1	2.6	416.4	31.8	32.4	2.9	896.7	450.45	20.6	0.3
R201_3_2	766.5	59.5	23.2	4.0	759.2	41.9	34.2	2.1	883.0	213.49	20.9	0.5

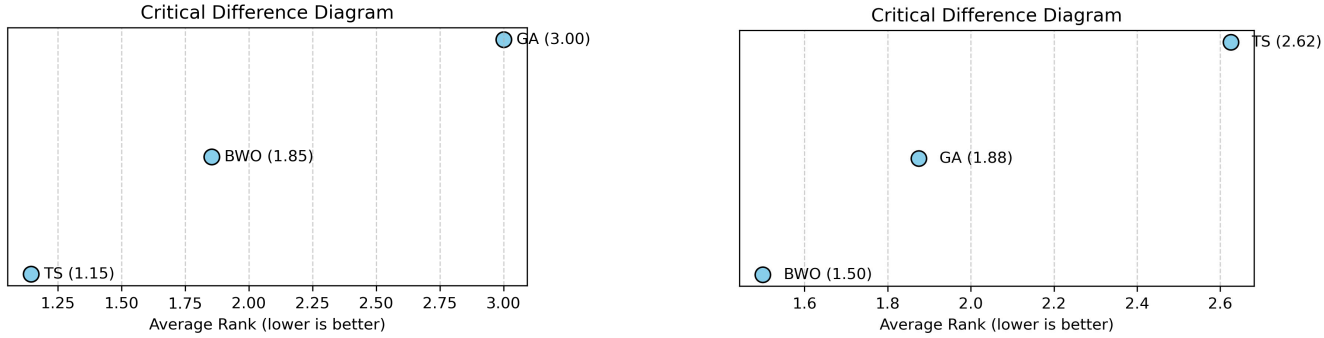


FIGURE 8. Performance of algorithm (a) Comparison CPU time and (b) Comparison fitness value.

TABLE 6. Post-hoc Nemenyi fitness value.

	BWO	GA	TS
BWO	1	0.3957	0.0003
GA	0.3957	1	0.0254
TS	0.0003	0.0254	1

TABLE 7. Post-hoc Nemenyi CPU time.

	BWO	GA	TS
BWO	1	0.0002	0.0376
GA	0.0002	1	0.00001
TS	0.0376	0.00001	1

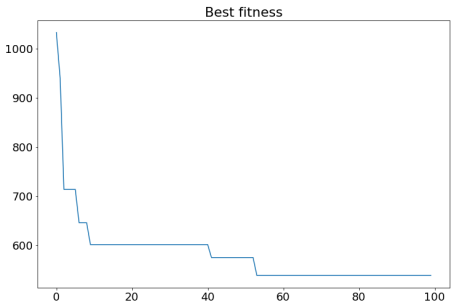


FIGURE 9. Convergence graph for the RC101_2_2 instance.

for supporting tactical and operational decision-making in last-mile logistics.

C. CASE STUDY

This section aims to analyze the BWO algorithm in a real-world context and determine its applicability in operational decisions. Notice that the real-case analysis distinguishes between B2B (business-to-business) and B2C (business-to-consumer) customers, each characterized by specific demand profiles, service durations, and time windows.

The dataset corresponds to twelve real operational days from June 2023, comprising more than 1,500 nodes,

TABLE 8. Instances of case study.

Instance	I	NC	NB	K	W
1000	10	10	5	8	26
2406	98	115	22	26	129
2706	108	118	61	39	141
1906	110	119	42	28	138
1706	117	132	24	30	152
2206	144	160	87	30	108
3006	158	191	39	36	176
2006	177	198	52	39	151
1006	178	202	102	32	128
2306	185	218	67	46	85
2906	215	245	104	23	176
806	235	268	119	36	158

including individual houses, apartments, or commercial buildings, served by a fleet of up to 50 vehicles. Figure 10 illustrates the spatial distribution of the nodes across the city. It is essential to note that some nodes include multiple customers, such as office towers, condominiums, or business centers, meaning that the total number of customers may exceed the number of physical stops. These days were selected because they represent periods of regular demand, with approximately 10% of all stops corresponding to multi-customer buildings.

The parameters for each instance are shown in Table 8, which includes the number of stopping nodes (I), e.g., houses or buildings, that day, the number of customers (NC), the number of businesses (NB), the number of vehicles (K), and the required number of workers (W). The company provided the data of more than 100,000 deliveries, and based on this information, we conducted an empirical estimation of demand by customer type: B2B customers typically receive between 3 and 10 packages, while B2C customers generally receive one package. This information is also confirmed by the works in [59] and [60]. Thus, each vehicle was assigned a capacity of 50 packages, both aligning with the company's operational worker experience. These assumptions were later reviewed and validated by the company.

Service times for each delivery location were set between 5 and 15 minutes for B2C customers and between 9 and 25 minutes for B2B customers, as informed by the study of Basso et al. in [61]. These times were subsequently adapted based on time dependency and delivery personnel, using the method proposed by Paredes-Belmar et al. in [8], which defines the role of delivery worker combinations described in Table 3, where the maximum capacity of each worker tupe was randomly determined based on the number of vehicles available per day. After these adjustments, the resulting service times were reviewed and validated with the delivery company.

The travel time for a trip can differ depending on the time of day due to vehicular congestion within dense urban areas. We used the study of [29] to determine vehicle speeds and travel times in Santiago, Chile. This study, conducted by the Logistics and Transport Center of Universidad Andres Bello, processed GPS data collected from over 20,000 cargo vehicles, reported every 30 seconds, during a four-year period (2016-2019). For our study, each node in the case study was assigned to a corresponding network node, and the travel time was assigned to each node accordingly. Additionally, we consider each interval of time to be one hour.

We divided the workday into ten one-hour time intervals, ranging from 8:00 AM to 6:00 PM. We tested various worker capacities on different days using the BWO metaheuristic, employing parameters similar to those used for large instances. We maintained consistent values for np, cr, nm, and I_{max}, with a population size ($NPOP$) of 100 individuals for these instances. It is important to note that for the tuned version of the novel BWO in these cases, a larger $NPOP$ is required, which significantly increases CPU time. These comparisons are available in B. The BWO was executed ten times, and we recorded the minimum, maximum, and average values of fitness, CPU time, number of vehicles, and number of workers. These results are presented in Table 9.

In all scenarios, the travel time is less than the 10 business working hours of the logistics company. It is also evident that the maximum execution time for this problem with these parameters is 30 minutes, which is advantageous for a company when making daily decisions in less than an hour at the beginning of the day. Additionally, it can be observed that both the number of vehicles and the number of workers used in all cases are less than those available to the company, which is also favorable. This demonstrates that the algorithm can be efficiently applied to real-life route planning in terms of time and resources. While optimality gaps cannot be computed for real-world instances, solution quality is evaluated via repeated runs and comparison.

1) INSTANCE 806 ANALYSIS

In this subsection, we focus on Instance 806 for a detailed analysis of its results, as this instance features a higher number of customers, particularly business customers. Table 10 presents the travel time in hours, the CPU time in minutes, the crew, and the vehicles required for each algorithm's implementation. Additionally, it compares travel time results in hours using the same routes with only one MFDP per vehicle. The results in Table 9 were executed 10 times for each instance. It is worth noting that all instances exhibit similar behavior to this one. The purpose of analyzing this specific instance is to provide a more detailed view of the results.

Note that the algorithm does not always provide the same objective value for large instances. Also, more resources (vehicles and crews) are required for shorter travel times. This is due to the dependence on the service time related to the number of workers used. This paper aims to minimize travel times, but with increased resource requirements and, consequently, a higher monetary cost for the company, which should be considered in decision-making.

We can observe a time difference of approximately two hours when using only one MFDP per vehicle on the same routes obtained with BWO. The option with only one MFDP

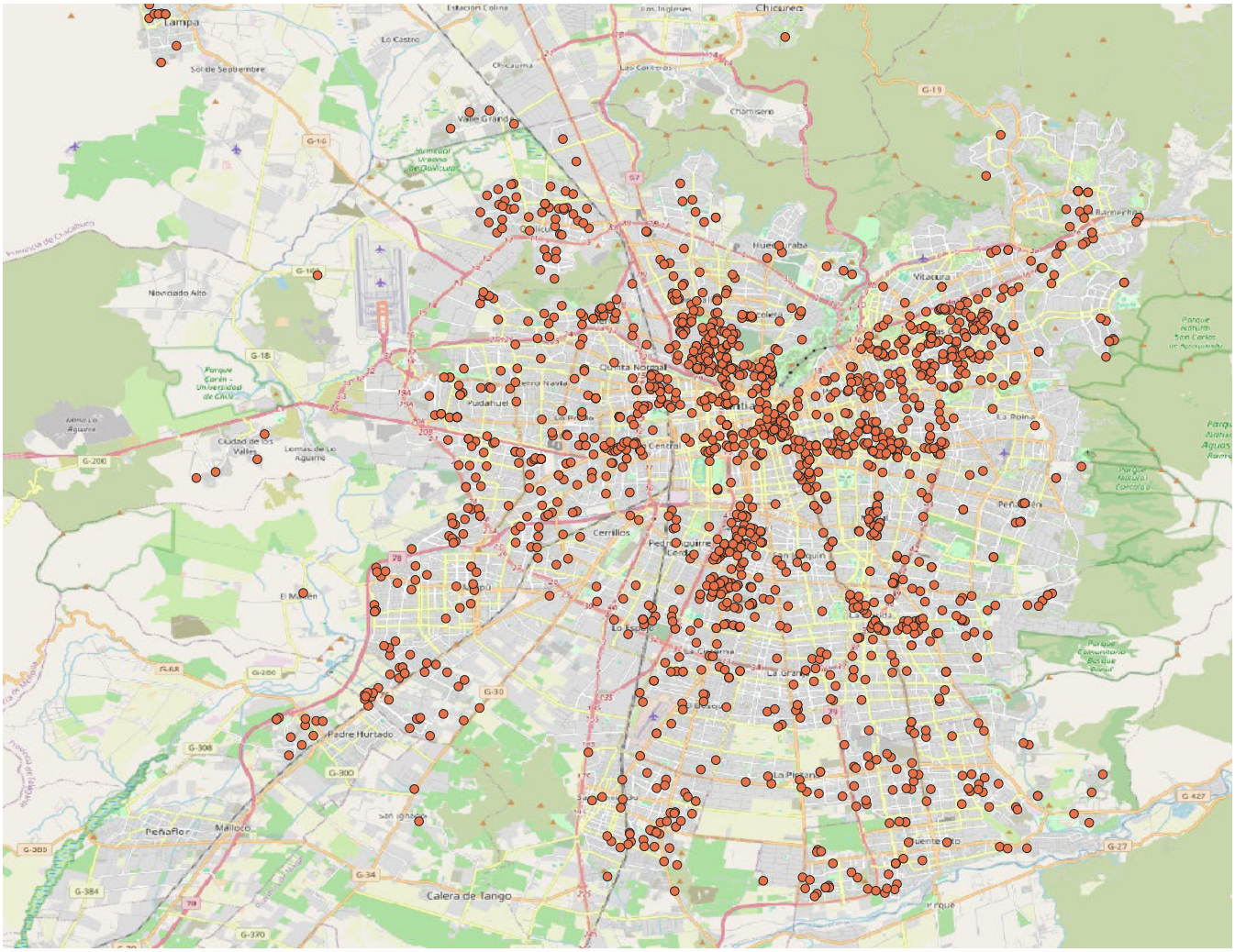


FIGURE 10. Nodes in case study in metropolitan region, chile.

TABLE 9. Case study results.

Instance	TT (h)			CPU (min)			Vehicles used			Workers used		
	Min	Average	Max	Min	Average	Max	Min	Average	Max	Min	Average	Max
1000	1.0	1.1	1.3	1.2	1.5	1.8	7	7	8	14	14	15
2406	4.3	4.6	5.1	9.8	12.7	14.8	22	25	25	51	71	78
2706	3.9	4.0	4.1	11.5	14.7	17.7	31	34	34	69	89	91
1906	4.7	5.0	5.5	11.4	14.1	17.2	23	26	28	28	64	88
1706	4.8	4.9	5.1	12.4	15.7	18.4	25	27	28	28	42	92
2206	5.8	6.3	6.6	14.9	20.4	24.1	25	27	29	76	86	89
3006	5.5	6.0	6.3	16.6	21.0	24.1	29	32	34	44	95	107
2006	6.9	7.2	7.7	19.8	24.0	27.2	27	28	31	90	97	104
1006	6.0	6.3	6.6	19.6	24.7	27.7	31	33	37	105	116	135
2306	5.4	6.3	6.6	20.9	25.8	29.3	36	38	41	43	105	131
2906	6.8	7.0	7.2	24.7	29.7	34.6	37	38	40	93	120	135
806	7.8	8.7	9.5	30.1	36.0	41.7	31	34	36	73	97	112

requires the most service time, as it considers the time required for one person to complete all tasks, including operations such as loading and unloading for each node. Additionally, it is necessary to find a parking space in this option.

Table 11 summarizes the travel time and crew composition for each vehicle in the fourth implementation of the case study. The table reports, for every K, the total travel time (TT), the number of drivers ($\#Dr$), and MFDPs ($\#MFDP$), as well as the travel time if only one MFDP were assigned to that same route. When a vehicle operates for eight hours to make deliveries, it utilizes its maximum cabin capacity of five workers. For instance, vehicle 36 employs five workers for long-distance travel, while vehicle 19 only requires one

TABLE 10. Results per implementation for the instance 806.

	Results BWO				One MFDP per vehicle
Implementation	TT (h)	CPU (min)	Vehicles used	Workers used	TT (h)
1	8.4	31.5	32	79	9.4
2	9.5	32.9	34	105	10.1
3	9.1	39.3	31	73	10.1
4	8.0	39.1	36	104	8.9
5	8.8	36.0	34	105	9.7
6	8.8	30.1	34	105	9.7
7	7.8	35.2	33	112	8.4
8	9.1	37.4	31	73	10.1
9	8.6	36.7	36	106	9.1
10	8.7	41.7	34	105	9.4
Average	8.7	36	34	97	9.5
Min	7.8	30.1	31	73	8.4
Max	9.5	41.7	36	112	10.1

worker due to its shorter travel time. This instance requires 18 drivers, 71 delivery personnel, and 18 multi-functional delivery personnel, amounting to 107 workers.

When comparing travel times per vehicle with only one multi-functional delivery person (MFDP) in table 11, we observe that most vehicles take longer, except in cases where one MFDP is involved. On average, there is a time difference of 0.23 hours between routes for the vehicles with and without the additional MFDP. The maximum observed difference is one hour, which indicates that more nodes are visited simultaneously when more personnel are present in the vehicle. It is important to note that when a vehicle operates with only one MFDP, it still visits the same nodes and follows the same route, but with a total of 36 workers on board.

In this particular instance, the nodes are scattered throughout the Metropolitan Region of Santiago, Chile, as shown in Figure 11, where each vehicle delivers to a maximum of 12 nodes due to the number of vehicles available.

In summary, this case study confirms the effectiveness of the novel BWO metaheuristic in a real-world instance, which features diverse customer types and specific service constraints. The algorithm delivers high-quality and feasible solutions within a reasonable timeframe, ensuring the efficient use of available resources and often requiring fewer vehicles and workers than initially planned.

While we cannot verify optimality due to the complexities of real-world data, the quality of the BWO solutions is evident from consistent results across multiple runs and a comparative analysis with constrained configurations (e.g., assigning one MFDP per vehicle). These findings suggest that the model and algorithm can be successfully integrated into daily planning processes, providing logistics operators with a reliable tool for balancing service levels, travel time, and operational costs.

Furthermore, the ability to distinguish between B2B and B2C profiles, incorporate realistic building-level service times, and consider time-dependent travel conditions enhances the accuracy of our methodology and decision-making capacity. This highlights the importance of integrating route optimization with detailed workforce assignment, particularly in complex urban environments like Santiago, Chile.

TABLE 11. Travel time and crew composition per vehicle.

K	TT (h)	#Dr	#DP	# MFDP	TT (h); One MFDP
1	8.0	1	1	0	8.3
2	8.0	1	1	0	8.9
3	1.0	1	1	0	1.2
4	4.1	1	1	0	4.7
5	7.6	1	2	0	8.6
6	2.4	1	2	0	2.4
7	4.7	1	2	0	4.7
8	3.4	1	2	0	3.5
9	6.6	1	3	0	6.6
10	1.3	1	3	0	1.3
11	3.8	1	3	0	4.0
12	6.8	1	3	0	7.3
13	7.0	1	3	0	7.1
14	1.3	1	3	0	1.4
15	4.9	1	3	0	5.0
16	7.2	1	3	0	7.2
17	7.5	1	4	0	7.8
18	4.6	1	4	0	5.0
19	6.6	0	0	1	6.6
20	1.0	0	0	1	1.0
21	0.7	0	0	1	0.7
22	1.3	0	0	1	1.3
23	6.1	0	0	1	6.1
24	0.8	0	0	1	0.8
25	5.2	0	0	1	5.2
26	1.7	0	1	1	1.7
27	7.7	0	1	1	7.7
28	1.4	0	2	1	1.5
29	4.2	0	2	1	4.8
30	2.9	0	2	1	3.2
31	2.3	0	2	1	2.8
32	7.7	0	2	1	7.8
33	1.3	0	3	1	1.4
34	5.0	0	3	1	5.9
35	6.0	0	3	1	6.1
36	8.0	0	3	1	8.9

D. MANAGERIAL IMPLICATIONS

The TD-VRPFCA presents a comprehensive framework for optimizing urban delivery operations by considering time-dependent delivery, service times, vehicle fleet, and crew assignment. Below are detailed managerial implications for decision-makers in last-mile logistics:

- Urban delivery routes are highly sensitive to time-of-day factors such as traffic congestion, availability of parking, and building access. Traditional route planning methods often fail to account for these time-sensitive variables,

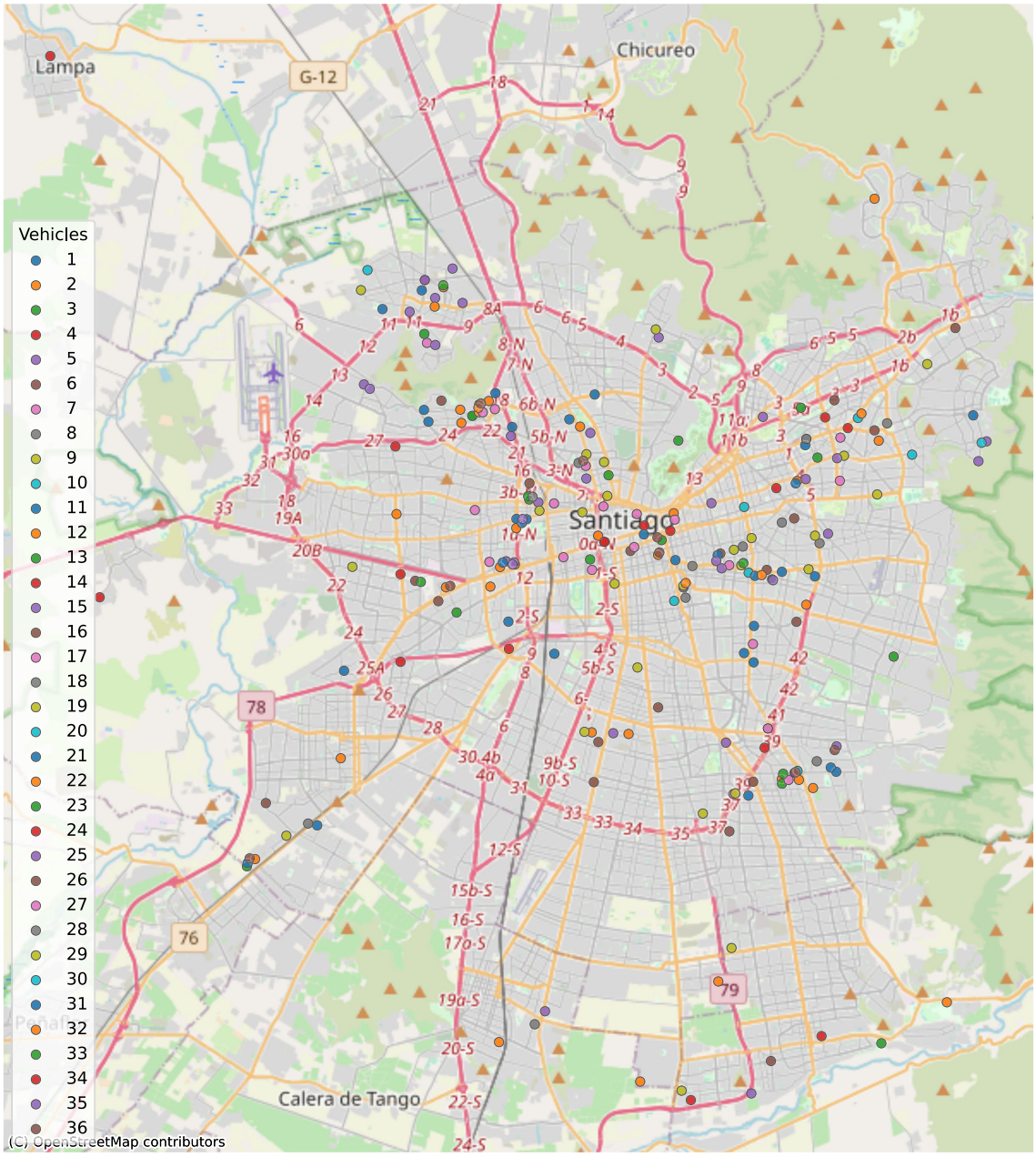


FIGURE 11. Instance 806.

leading to inefficiencies. In a city like Santiago, Chile, a delivery that would take 15 minutes at 11 a.m. might take 45 minutes at 5 p.m. due to peak hours on the streets. Moreover, an office building with 20 floors may have slower elevator service during peak hours (morning, lunchtime, and afternoon) because many of its workers are using it. The TD-VRPFCA integrates time-dependent travel times and time-dependent service times, which are crucial in urban environments where traffic and service conditions fluctuate throughout the day. This enables logistics managers to create delivery plans that accurately reflect real-time traffic in urban areas.

- The TD-VRPFCA incorporates the crew assignment for delivery crews based on the requirements of deliveries and time of day. In this context, the results highlight that assigning flexible crews containing delivery personnel and drivers significantly reduces overall delivery times. In contrast, restricting vehicles to only one crew member leads to up to two additional hours of operations per

TABLE 12. Average travel times and CPU for different parameter combinations.

np	*cr*	*nm*	*npop*	i_{max}	Average Travel Time				Avg. CPU (s)
					RC101_1_1	RC201_1_1	R101_1_1	R201_1_1	
0.4	0.5	0.05	20	50	468.3	472.21	433.66	264.05	10.14
0.4	0.5	0.05	20	100	386.3	442.30	568.37	259.85	19.81
0.4	0.5	0.05	50	50	394.7	372.87	551.42	254.24	64.07
0.4	0.5	0.05	50	100	377.1	381.45	523.13	234.90	125.33
0.4	0.5	0.05	100	50	368.2	369.22	531.74	227.59	265.23
0.4	0.5	0.05	100	100	345.9	341.28	520.09	214.07	530.52
0.4	0.5	0.10	20	50	461.8	417.19	569.04	265.29	10.68
0.4	0.5	0.10	20	100	421.0	380.47	512.34	256.59	20.74
0.4	0.5	0.10	50	50	391.6	414.57	535.47	241.93	67.86
0.4	0.5	0.10	50	100	295.3	328.94	532.28	205.86	132.72
0.4	0.5	0.10	100	50	354.4	334.58	527.32	203.63	279.23
0.4	0.5	0.10	100	100	321.5	333.15	502.69	196.68	548.03
0.4	1.0	0.05	20	50	469.2	467.99	594.33	282.61	10.28
0.4	1.0	0.05	20	100	419.4	444.71	534.95	309.43	20.19
0.4	1.0	0.05	50	50	405.6	402.66	553.20	266.65	65.60
0.4	1.0	0.05	50	100	338.8	363.34	528.24	217.38	128.05
0.4	1.0	0.05	100	50	356.5	343.51	532.06	231.93	270.27
0.4	1.0	0.05	100	100	340.7	364.18	508.42	209.34	528.82
0.4	1.0	0.10	20	50	456.6	452.09	556.64	272.01	10.93
0.4	1.0	0.10	20	100	442.1	408.43	540.01	259.87	21.21
0.4	1.0	0.10	50	50	374.8	374.75	545.39	253.63	68.16
0.4	1.0	0.10	50	100	356.6	345.12	519.25	206.81	133.12
0.4	1.0	0.10	100	50	315.7	337.24	429.27	232.52	278.71
0.4	1.0	0.10	100	100	280.9	313.28	484.75	188.48	547.94

TABLE 13. Performance of the algorithm for different NPOP sizes across instances.

Instance	NPOP = 50		NPOP = 100		NPOP = 200		NPOP = 300	
	TT (h)	CPU (min)	TT (h)	CPU (min)	TT (h)	CPU (min)	TT (h)	CPU (min)
1000	1.2	0.3	1.1	1.5	1.2	5.8	1.1	15.5
2406	4.8	3.3	4.6	12.7	4.6	52.7	4.4	129.4
2706	4.2	3.7	4.0	14.7	4.6	62.2	4.0	151.2
1906	5.2	3.6	5.0	14.1	4.8	58.2	4.8	145.8
1706	5.3	3.8	4.9	15.7	4.9	64.4	4.5	141.9
2206	6.4	3.9	6.3	20.4	6.2	81.5	5.9	188.4
3006	6.2	5.9	6.0	21.0	5.9	85.8	5.8	149.4
2006	7.5	6.6	7.2	24.0	6.9	81.4	7.0	109.8
1006	6.6	6.8	6.3	24.7	6.3	95.2	6.1	132.9
2306	6.6	6.8	6.3	25.8	5.6	85.5	6.3	136.9
2906	7.6	8.2	7.0	29.7	7.1	132.5	7.0	137.8
806	9.4	9.9	8.7	36.0	8.8	140.8	8.9	165.8

route. Thus, a manager can choose between resource-intensive, fast deliveries with more vehicles or larger crews, or slower, cheaper operations. Consequently, the TD-VRPFCA provides decision-makers with a structured way to balance workforce and service quality, depending on the company's operational priorities (e.g., minimizing overtime, delivery personnel, or meeting tight delivery windows).

- The case study proves that the proposed approach often requires fewer vehicles and workers than those currently available to the company, while still meeting delivery demand within standard working hours. This indicates a potential for cost savings through fleet downsizing or worker reassignment to other tasks.
- The Black Widow Optimization (BWO) metaheuristic, as presented in the article, demonstrates effectiveness for both small- and large-scale delivery problems. In a real context, delivery companies usually have little time (e.g., one or two hours before delivery) for route planning due to the nature of operational decisions. Thus, quick and good-quality solutions are required in a short time. Consequently, the BWO metaheuristic is very useful for decision-making.
- The Novel BWO algorithm consistently produced stable solutions in all instances tested, including real-case data. The low standard deviation in fitness and CPU time across multiple runs confirms its robustness, a critical property for deployment in operational planning environments with uncertainty and time pressure.
- The TD-VRPFCA distinguishes between street-level and in-building deliveries, incorporating time-dependent service times to account for the complexities of navigating buildings. Many companies fail to distinguish between these factors, resulting in inaccurate time estimates and delays. Our results suggest establishing partnerships with large buildings to improve delivery logistics. For example, dedicated delivery parking zones or service elevators could significantly reduce delivery times.
- Beyond daily planning, the TD-VRPFCA can provide helpful information for tactical decisions. It enables companies to forecast workforce requirements under various demand patterns, assess the operational impact of possible regulatory changes, and evaluate the benefits of labor models. Furthermore, it provides a basis for evaluating environmental impacts, such as reductions

in fuel consumption and emissions, thereby supporting sustainability purposes.

- The comparative analysis among the three metaheuristics—BWO, GA, and TS—also provides managerial insights. The BWO consistently achieved comparable or superior routing solutions with substantially lower computational times, allowing planners to evaluate multiple delivery scenarios more efficiently. Its balance between diversification and intensification makes it especially suitable for time-dependent and heterogeneous crew conditions. In contrast, GA tends to converge faster but may produce less balanced crew allocations, while TS offers stability at the expense of longer runtimes. Therefore, the BWO can be considered a more flexible and efficient optimization framework for supporting real-time decision-making in last-mile delivery operations.

V. CONCLUSION

We introduced the Time-Dependent Vehicle Routing Problem with Fleet and Crew Assignment (TD-VRPFCA), which aims to find efficient delivery routes for vehicles in urban areas. This includes serving customers at street level and within buildings while minimizing distribution times. In addition, we consider the personnel traveling in the vehicles and incorporate time-dependency to estimate distribution times accurately.

To solve small-size instances of the TD-VRPFCA, we formulated a Mixed-Integer Linear Programming (MILP) model. We have developed a Black Widow Optimization (BWO) metaheuristic for larger real-world instances. Both methods effectively handle the complexity and variability of travel and service times in urban environments.

We have conducted extensive computational experiments using test instances and a real-world case study from a Chilean delivery company. Our results show that our approach significantly optimizes delivery routes and times, demonstrating its efficiency.

The findings underscore the significance of precise time estimations in delivery operations, which can significantly impact overall efficiency. By differentiating between street-level and in-building deliveries, the model addresses a critical gap in current logistics planning that is often overlooked in traditional routing problems. Integrating multi-functional delivery persons who can drive and assist in deliveries provides a practical solution to the challenges posed by urban delivery environments.

Our study provides new perspectives and tools to address complex urban logistics and in-building deliveries, offering practical and efficient solutions for the distribution industry.

Implementing the BWO metaheuristic proved effective in managing the complexity of large instances, showcasing its potential for broader applications in logistics optimization. Compared to other metaheuristic approaches, the BWO's performance in handling time-dependent variables and crew assignments makes a significant contribution to the study area of last-mile delivery.

Overall, this study not only demonstrates the effectiveness of the TD-VRPFCA model and the BWO algorithm but also highlights their potential as decision-support tools for real-world logistics. Beyond operational planning, the model enables the simulation of alternative crew strategies, quantification of performance trade-offs, and integration of real-world data such as GPS mobility patterns and building-level logistics constraints. This makes the framework applicable for both tactical decision-making and strategic policy design in urban distribution systems. The ability to distinguish between customer types and adapt delivery strategies accordingly also contributes to more resilient and adaptive last-mile solutions.

Future research can build on this study by exploring several directions:

1) Extended crew configurations: Expanding the range of crew configurations and examining the impact of varying worker skill levels on delivery efficiency can provide deeper insights into optimal crew management.
2) Environmental impact analysis: Assessing the environmental benefits of optimized delivery routes, such as reduced emissions and fuel consumption, can support sustainable logistics practices.
3) Economic impact studies: Conducting comprehensive economic analyses to quantify cost savings and productivity gains achieved through optimized routing and crew assignments can provide valuable information for logistics companies and urban planners.

By addressing these areas, future research can continue to enhance the efficiency and sustainability of urban delivery systems, ultimately contributing to more resilient and adaptive logistics networks.

APPENDIX A
SUMMARY RESULTS LARGE INSTANCES FOR FRIEDMAN TEST

In the following table 12, we summarize the results using different combinations of parameters for metaheuristic in Large Instances. These results were used for the Friedman test. The best combination based on ranks was np: 0.6, cr: 0.5, nm: 0.1, $NPOP$: 20, I_{max}: 100.

APPENDIX B
DIFFERENCE CPU TIME FOR NPOP

The BWO metaheuristic was tested with different parameters, specifically the size of the initial population, to observe the BWO's efficiency with relatively small parameters for the number of nodes being evaluated. The size of the initial population enables the algorithm to search for a better solution with a large set of solutions that it can evaluate. Having more solutions at the beginning makes it less likely for the algorithm to get stuck in a local optimum and allows it to move to a global optimum more easily. For

this reason, the algorithm is executed 10 times with initial populations (*NPOP*) of size 200, 300, 400, and 500, and then compared with the initial size of 100 to determine whether the differences in the values are significant or not according to the *U-test*. Table 13 presents the average value of the results.

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REFERENCES

[1] P. Toth and D. Vigo, *Vehicle Routing: Problems, Methods, and Applications*. Philadelphia, PA, USA: SIAM, 2014.

[2] D. Cattaruzza, N. Absi, D. Feillet, and J. González-Feliu, "Vehicle routing problems for city logistics," *EURO J. Transp. Logistics*, vol. 6, no. 1, pp. 51–79, Mar. 2017.

[3] H. Á. R. Navarro, J. G. V. Vieira, and J. C. Fransoo, "Understanding urban logistics and consumer behavior in São Paulo City," *Case Stud. Transp. Policy*, vol. 9, no. 4, pp. 1630–1640, Dec. 2021.

[4] J. F. Ehmke, A. Steinert, and D. C. Mattfeld, "Advanced routing for city logistics service providers based on time-dependent travel times," *J. Comput. Sci.*, vol. 3, no. 4, pp. 193–205, Jul. 2012.

[5] S. Reed, A. M. Campbell, and B. W. Thomas, "Does parking matter? The impact of parking time on last-mile delivery optimization," *Transp. Res. E, Logistics Transp. Rev.*, vol. 181, Jan. 2024, Art. no. 103391.

[6] N. Ando and E. Taniguchi, "Travel time reliability in vehicle routing and scheduling with time windows," *Netw. Spatial Econ.*, vol. 6, nos. 3–4, pp. 293–311, Sep. 2006.

[7] V. Pureza, R. Morabito, and M. Reimann, "Vehicle routing with multiple deliverymen: Modeling and heuristic approaches for the VRPTW," *Eur. J. Oper. Res.*, vol. 218, no. 3, pp. 636–647, May 2012.

[8] G. Paredes-Belmar, G. Latorre-Núñez, and A. Bronfman, "Modeling and solving the time-dependent in-building delivery problem in last-mile logistics," *IEEE Access*, vol. 12, pp. 11276–11293, 2024.

[9] K. Braekers, K. Ramaekers, and I. Van Nieuwenhuyse, "The vehicle routing problem: State of the art classification and review," *Comput. Ind. Eng.*, vol. 99, pp. 300–313, Sep. 2016.

[10] S. Ichoua, M. Gendreau, and J.-Y. Potvin, "Vehicle dispatching with time-dependent travel times," *Eur. J. Oper. Res.*, vol. 144, no. 2, pp. 379–396, Jan. 2003.

[11] J. F. Ehmke, *Attended Home Delivery*. Boston, MA, USA: Springer, 2012.

[12] M. Gendreau, G. Ghiani, and E. Guerriero, "Time-dependent routing problems: A review," *Comput. Oper. Res.*, vol. 64, pp. 189–197, Dec. 2015.

[13] Y. Xiao and A. Konak, "The heterogeneous green vehicle routing and scheduling problem with time-varying traffic congestion," *Transp. Res. E, Logistics Transp. Rev.*, vol. 88, pp. 146–166, Apr. 2016.

[14] J. Blauth, S. Held, D. Müller, N. Schlomberg, V. Traub, T. Tröbst, and J. Vygen, "Vehicle routing with time-dependent travel times: Theory, practice, and benchmarks," *Discrete Optim.*, vol. 53, Aug. 2024, Art. no. 100848.

[15] B. Dosunmu, "Delivering London 2012: Transport demand forecasting," *Proc. Inst. Civil Eng. Transp.*, vol. 165, no. 4, pp. 257–266, Nov. 2012.

[16] S. Kershaw, "Delivering London 2012: Transport programme assurance," *Proc. Inst. Civil Eng. Transp.*, vol. 165, no. 4, pp. 241–248, Nov. 2012.

[17] H. Sumner, "Delivering London 2012: Transport legacy," *Proc. Inst. Civil Eng. Transp.*, vol. 165, no. 4, pp. 267–275, Nov. 2012.

[18] P. Munari and R. Morabito, "A branch-price-and-cut algorithm for the vehicle routing problem with time windows and multiple deliverymen," *TOP*, vol. 26, no. 3, pp. 437–464, Oct. 2018.

[19] A. Alvarez and P. Munari, "An exact hybrid method for the vehicle routing problem with time windows and multiple deliverymen," *Comput. Oper. Res.*, vol. 83, pp. 1–12, Jul. 2017.

[20] M.-X. Zhang, J.-Y. Wu, X. Wu, and Y.-J. Zheng, "Hybrid evolutionary optimization for takeaway order selection and delivery path planning utilizing habit data," *Complex Intell. Syst.*, vol. 8, no. 6, pp. 4425–4440, Dec. 2022.

[21] A. A. Kovacs, S. N. Parragh, K. F. Doerner, and R. F. Hartl, "Adaptive large neighborhood search for service technician routing and scheduling problems," *J. Scheduling*, vol. 15, no. 5, pp. 579–600, Oct. 2012.

[22] S. C. Ho and J. M. Y. Leung, "Solving a manpower scheduling problem for airline catering using metaheuristics," *Eur. J. Oper. Res.*, vol. 202, no. 3, pp. 903–921, May 2010.

[23] N. Cabrera, J.-F. Cordeau, and J. E. Mendoza, "The doubly open park-and-loop routing problem," *Comput. Oper. Res.*, vol. 143, Jul. 2022, Art. no. 105761.

[24] T. Le Colleter, D. Dumez, F. Lehuédé, and O. Péton, "Small and large neighborhood search for the park-and-loop routing problem with parking selection," *Eur. J. Oper. Res.*, vol. 308, no. 3, pp. 1233–1248, Aug. 2023.

[25] A. Martinez-Sykora, F. McLeod, C. Lamas-Fernandez, T. Bektaş, T. Cherrett, and J. Allen, "Optimised solutions to the last-mile delivery problem in London using a combination of walking and driving," *Ann. Oper. Res.*, vol. 295, no. 2, pp. 645–693, Dec. 2020.

[26] O. Gkalp and A. Uur, "Solving vehicle routing problem for multistorey buildings using iterated local search," *TURKISH J. Electr. Eng. Comput. Sci.*, vol. 27, no. 5, pp. 3516–3531, Sep. 2019.

[27] H. Kim, L. N. Boyle, and A. Goodchild, "Delivery process for an office building in the Seattle central business district," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2672, no. 9, pp. 173–183, Dec. 2018.

[28] M.-C. Chen, S. Yerasani, and M. K. Tiwari, "Solving a 3-dimensional vehicle routing problem with delivery options in city logistics using fast-neighborhood based crowding differential evolution algorithm," *J. Ambient Intell. Humanized Comput.*, vol. 14, no. 8, pp. 10389–10402, Aug. 2023.

[29] F. Menares, E. Montero, G. Paredes-Belmar, and A. Bronfman, "A bi-objective time-dependent vehicle routing problem with delivery failure probabilities," *Comput. Ind. Eng.*, vol. 185, Nov. 2023, Art. no. 109601.

[30] V. D. O. Ferreira and V. Pureza, "Some experiments with a savings heuristic and a Tabu search approach for the vehicle routing problem with multiple deliverymen," *Pesquisa Operacional*, vol. 32, no. 2, pp. 443–463, Jun. 2012.

[31] J. F. D. Souza Neto and V. Pureza, "Modeling and solving a rich vehicle routing problem for the delivery of goods in urban areas," *Pesquisa Operacional*, vol. 36, no. 3, pp. 421–446, Dec. 2016.

[32] J. De La Vega, P. Munari, and R. Morabito, "Exact approaches to the robust vehicle routing problem with time windows and multiple deliverymen," *Comput. Oper. Res.*, vol. 124, Dec. 2020, Art. no. 105062.

[33] E. Taniguchi and H. Shimamoto, "Intelligent transportation system based dynamic vehicle routing and scheduling with variable travel times," *Transp. Res. C, Emerg. Technol.*, vol. 12, nos. 3–4, pp. 235–250, Jun. 2004.

[34] Y. Kuo, "Using simulated annealing to minimize fuel consumption for the time-dependent vehicle routing problem," *Comput. Ind. Eng.*, vol. 59, no. 1, pp. 157–165, Aug. 2010.

[35] W. Maden, R. Eglese, and D. Black, "Vehicle routing and scheduling with time-varying data: A case study," *J. Oper. Res. Soc.*, vol. 61, no. 3, pp. 515–522, Mar. 2010.

[36] S. Kritzinger, K. F. Doerner, R. F. Hartl, G. Ÿ. Kiechle, H. Stadler, and S. S. Manohar, "Using traffic information for time-dependent vehicle routing," *Proc. Social Behav. Sci.*, vol. 39, pp. 217–229, Jan. 2012.

[37] M. Çimen and M. Soysal, "Time-dependent green vehicle routing problem with stochastic vehicle speeds: An approximate dynamic programming algorithm," *Transp. Res. D, Transp. Environ.*, vol. 54, pp. 82–98, Jul. 2017.

[38] S. Allahyari, S. Yaghoubi, and T. Van Woensel, "The secure time-dependent vehicle routing problem with uncertain demands," *Comput. Oper. Res.*, vol. 131, Jul. 2021, Art. no. 105253.

[39] N. Ouertani, H. Ben-Romdhane, S. Krichen, and I. Nouaouri, "A vector evaluated evolutionary algorithm with exploitation reinforcement for the dynamic pollution routing problem," *J. Combinat. Optim.*, vol. 44, no. 2, pp. 1011–1038, Sep. 2022.

[40] S. I. Nyako, D. Tayachi, and M. Tagina, "Minimizing fuel consumption in the time-dependent VRP," in *Proc. 8th Int. Conf. Control, Decis. Inf. Technol. (CoDIT)*, May 2022, pp. 647–652.

[41] B. Naderi, M. A. Begen, G. S. Zaric, and V. Roshanaei, "A novel and efficient exact technique for integrated staffing, assignment, routing, and scheduling of home care services under uncertainty," *Omega*, vol. 116, Apr. 2023, Art. no. 102805.

[42] M. Rahbari, A. Arshadi Khamseh, Y. Sadati-Keneti, and M. J. Jafari, "A risk-based green location-inventory-routing problem for hazardous materials: NSGA II, MOSA, and multi-objective black widow optimization," *Environ., Develop. Sustainability*, vol. 24, no. 2, pp. 2804–2840, Feb. 2022.

[43] V. Hayyolalam and A. A. P. Kazem, "Black widow optimization algorithm: A novel meta-heuristic approach for solving engineering optimization problems," *Eng. Appl. Artif. Intell.*, vol. 87, Jan. 2020, Art. no. 103249.

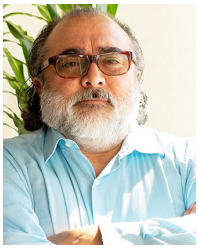
[44] P. Figueroa-Torrez, O. Durán, B. Crawford, and F. Cisternas-Caneo, "A binary black widow optimization algorithm for addressing the cell formation problem involving alternative routes and machine reliability," *Mathematics*, vol. 11, no. 16, p. 3475, Aug. 2023.
[45] M. Becerra-Rozas, J. Lemus-Romani, F. Cisternas-Caneo, B. Crawford, R. Soto, G. Astorga, C. Castro, and J. García, "Continuous metaheuristics for binary optimization problems: An updated systematic literature review," *Mathematics*, vol. 11, no. 1, p. 129, Dec. 2022.
[46] M. Pinedo, "Scheduling: Theory, algorithms, and systems," *Scheduling, Theory*, vol. 28, no. 8, pp. 695–697, Jan. 1996.
[47] G. Desaulniers, O. B. G. Madsen, S. Røpke, P. Toth, and D. Vigo, "Chapter 5: The vehicle routing problem with time windows," in *Vehicle Routing: Problems, and Applications*. Philadelphia, PA, USA: SIAM, 2014, ch. 5, pp. 119–159.
[48] M. Shehab, M. K. Y. Shambour, M. A. Abu Hashem, H. A. Al Hamad, F. Shannaq, M. Mizher, G. Jaradat, M. S. Daoud, and L. Abualigah, "A survey and recent advances in black widow optimization: Variants and applications," *Neural Comput. Appl.*, vol. 36, pp. 1–21, Mar. 2024.
[49] A. M. Jabbar and K. R. Ku-Mahamud, "Hybrid black widow optimization and variable neighborhood descent algorithm for traveling salesman problem," *Int. J. Systematic Innov.*, vol. 6, no. 5, pp. 32–43, 2021.
[50] Y. Fu, Y. Hou, Z. Chen, X. Pu, K. Gao, and A. Sadollah, "Modelling and scheduling integration of distributed production and distribution problems via black widow optimization," *Swarm Evol. Comput.*, vol. 68, Feb. 2022, Art. no. 101015.
[51] C. Meng, Z. Feng, D. Zhao, X. Li, J. Yu, and L. Yang, "A fuzzy scheduling method for pipeline processing in shipyards incorporating the black widow optimization algorithm," *Appl. Sci.*, vol. 14, no. 13, p. 5639, Jun. 2024.
[52] K. Cherrat and R. M. Essaid, "Black widow's new approach to tackle the traveling salesman problem," *J. Theor. Appl. Inf. Technol.*, vol. 102, no. 1, pp. 1–9, 2024.
[53] J. Allen, T. Cherrett, M. Piecyk, and M. Piotrowska, "The logistics of parcel delivery: Current operations and challenges facing the U.K. market," in *Urban Logistic: Management, Policy and Innovation in a Rapidly Changing Environment*, M. Browne, S. Behrends, J. Woxenius, G. Giuliano, and J. Holguin-Veras, Eds., London, U.K.: Kogan, 2019, pp. 141–166.
[54] S. R. Balseiro, I. Loiseau, and J. Ramonet, "An ant colony algorithm hybridized with insertion heuristics for the time dependent vehicle routing problem with time windows," *Comput. Oper. Res.*, vol. 38, no. 6, pp. 954–966, Jun. 2011.
[55] M. M. Solomon, "Algorithms for the vehicle routing and scheduling problems with time window constraints," *Oper. Res.*, vol. 35, no. 2, pp. 254–265, Apr. 1987.
[56] M. Gendreau and R. Seguin, "A Tabu search heuristic for the vehicle routing problem," *Oper. Res.*, vol. 44, no. 3, pp. 1276–1290, 1996.
[57] N. Labadie, C. Prins, and C. Prodhon, *Metaheuristics for Vehicle Routing Problems*. Hoboken, NJ, USA: Wiley, 2016.
[58] Q. Ni and Y. Tang, "A bibliometric visualized analysis and classification of vehicle routing problem research," *Sustainability*, vol. 15, no. 9, p. 7394, Apr. 2023.
[59] K. Aljohani and R. G. Thompson, "An examination of last mile delivery practices of freight carriers servicing business receivers in inner-city areas," *Sustainability*, vol. 12, no. 7, p. 2837, Apr. 2020.
[60] M. Schnieder, C. Hinde, and A. West, "Land efficient mobility: Evaluation of autonomous last mile delivery concepts in London," *Int. J. Environ. Res. Public Health*, vol. 19, no. 16, p. 10290, Aug. 2022.
[61] F. Basso, M. Núñez, G. Paredes-Belmar, R. Pezoa, and M. Varas, "Estimation of stops of last-mile delivery vehicles: An application in the food industry in the city of Santiago de Chile," *J. Transp. Geography*, vol. 116, Apr. 2024, Art. no. 103854.



ANTONIA ILABACA received the degree in industrial engineering and the M.Sc. degree in engineering sciences from Universidad Andés Bello, Viña del Mar, Chile, in 2019 and 2021, respectively. She is currently pursuing the Ph.D. degree in industrial engineering with Pontificia Universidad Católica de Valparaíso, Chile. Her doctoral research focuses on vehicle routing problems and fleet and crew assignment. Her work combines exact methods and metaheuristics to address logistics and transportation challenges. She has collaborated with both academic institutions and industry partners. Her research contributions have been presented at international conferences and she has teaching experience in operations research, logistics, and supply chain management at the undergraduate level.



GERMÁN PAREDES-BELMAR received the degree in industrial engineer from the Universidad del Bío-Bío, Concepción, Chile, in 2008, and the M.Sc. and Ph.D. degrees in engineering sciences from Pontificia Universidad Católica de Chile, Chile, in 2016. From 2014 to 2020, he was an Assistant Professor with the Department of Engineering Sciences, Universidad Andres Bello, Santiago, Chile. Since 2021, he has been an Assistant Professor with the School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Chile. Currently, he is Leading the FONDECYT Chilean Research Grant 1250602. His research interests include vehicle routing, network design, city logistics, and related applications.



BRODERICK CRAWFORD received the Ph.D. degree in computer science from Universidad Técnica Federico Santa María of Valparaíso, Chile, in 2011. He is currently a Full Professor of the Computer Science Department, Pontificia Universidad Católica de Valparaíso, Chile. In this context, he has published more than 300 scientific papers in different international conferences and journals, some of them top ranked in *Computer Science*, *Operational Research*, and *Artificial Intelligence*. Most of these papers are based on the resolution of benchmark and real-world optimization problems. His research interests include mainly combinatorial optimization, metaheuristics, global optimization, and autonomous search.